Cover Page

Standard: BAC5004-1   Rev: (R)   15-May-2008

1 SCOPE

NOTE: Incorporated PSDs: 6-64
Cancelled PSDs: None

This specification and BAC5004 describe the procedures for the installation and inspection of non-fluid-tight solid rivets as specified on the applicable Engineering drawings, where the stack up materials are metallic, except for solid rivets installed in composites, thermoplastic, and titanium structure. The composite and thermoplastic material, titanium structure, and fluid-tight structure are found in BAC5063, BAC5058, and BAC5047, respectively. Refer to BAC0001PREF for guidance on use and maintenance of Boeing process specifications and Boeing process specification departures.

2 CLASSIFICATION

Not applicable to this specification.

3 REFERENCES

The current issue of the following documents shall be considered a part of this specification to the extent indicated herein.

BAC5004	-	Installation of Permanent Fasteners
BAC5010	-	Application of Adhesives
BAC5034	-	Temporary Protection of Production Materials, Parts, and Assemblies
BAC5047	-	Fluid Tight Fastener Installation
BAC5049	-	Dimpling and Countersinking
BAC5058	-	Riveting in Titanium
BAC5063	-	Fastener Installation in Composite Structures
BAC5300	-	Forming, Straightening and Fitting Metal Parts
BAC5602	-	Heat Treatment of Aluminum Alloys
BAC5750	-	Solvent Cleaning
BSS7022	-	Procedures and Equipment for Measurement of Fastener Holes
BSS7286	-	Statistical Process Control of Designated Engineering Characteristics
D6-56617	-	Certification of Automatic Hydraulic-Squeeze Fastening Equipment
65-76487	-	Interference Coupon Drawing

4 CONTENTS

Section	Title	Page
1	SCOPE	1
2	CLASSIFICATION	1
3	REFERENCES	1
4	CONTENTS	2
5	MATERIALS CONTROL	6
6	FACILITIES CONTROL	6
6.1	CERTIFICATION OF EQUIPMENT	6
6.1.1	HYDRAULIC-SQUEEZE EQUIPMENT FOR BACR15FV, NAS1097, BACR15CE, AND BACR15GF RIVETS	6
7	DEFINITIONS	6
8	MANUFACTURING CONTROL	6
8.1	GENERAL	6
8.1.1	RIVET INSTALLATION	7
8.1.2	HOLES AND COUNTERSINKING	7
8.1.3	HOLE LOCATION	16
8.1.4	RIVET LENGTH	16
8.1.5	ALLOYS AND HEAT TREATMENT	19
8.1.6	FLUSH RIVETING	20
8.1.7	DRIVEN BUTTONS	21
8.1.8	TEMPORARY RIVET RETENTION	21
8.1.9	REWORK	21
8.2	IN-PROCESS VERIFICATION	25
8.3	STATISTICAL PROCESS CONTROL (SPC) OF RIVET INSTALLATIONS	26
8.4	PERSONNEL REQUIREMENTS	26
9	MAINTENANCE CONTROL	26
10	QUALITY CONTROL	26
11	REQUIREMENTS	28
11.1	FLUSHNESS	28
11.1.1	HEAD FLUSHNESS	28
11.1.2	SHAVING	29

4 CONTENTS (Continued)

Section	Title	Page
11.2	DRIVEN BUTTONS AND MANUFACTURED HEADS	29
11.2.1	STANDARD RIVET DRIVEN BUTTON DIMENSIONS	29
11.2.2	MANUFACTURED HEADS	34
11.2.3	DIAGONAL CRACKS	35
11.2.4	VERTICAL CRACKS	35
11.2.5	DEFORMED DRIVEN BUTTON	35
11.2.6	DRIVEN BUTTON IDENTIFICATION MARKING	37
11.3	GAPPED AND SHANKED RIVETS	37
11.3.1	GAPPED RIVETS	37
11.3.2	SHANKED RIVETS - PART SEPARATION AND SEALANT APPLICATION	41
12	TEST METHODS	42
13	QUALIFICATION	42

LIST OF FIGURES

Figure	Title	Page
FIGURE 1	RIVET HOLE LOCATION	16
FIGURE 2	RIVET LENGTH (L) AND GRIP (T) DIMENSIONS FOR NASM83459 (CHERRYBUCK) RIVETS	17
FIGURE 3	RIVET LENGTH (L) AND GRIP (T) DIMENSIONS FOR STANDARD ALUMINUM SOLID RIVETS	19
FIGURE 4	PARTIALLY EXPOSED COUNTERSINK	20
FIGURE 5	NASM14218 AND BACR15FV RIVET IN THICK SKIN	24
FIGURE 6	NASM14218 AND BACR15FV RIVET IN THIN SKIN	25
FIGURE 7	INSTALLATION OF EYELET-TYPE NUTPLATES	31
FIGURE 8	TUBULAR RIVET INSTALLATIONS	33
FIGURE 9	RADIAL CRACK ILLUSTRATION	33
FIGURE 10	ACCEPTABLE DEFORMATION OF THE MANUFACTURED HEAD	34
FIGURE 11	DEFORMATION NOT ACCEPTABLE	34
FIGURE 12	OVERSIZED RIVET DIE	34
FIGURE 13	CLINCHED RIVETS NOT ACCEPTABLE	36
FIGURE 14	BELL-SHAPED RIVETS NOT ACCEPTABLE	36
FIGURE 15	BACR15FV OR NASM14218 HEAD GAPS	38

4 **CONTENTS (Continued)**

Figure	Title	Page
FIGURE 16	GAP INSPECTION SHIM	38
FIGURE 17	PROTRUDING HEAD GAP CHECKING METHOD	39
FIGURE 18	PROTRUDING HEAD GAP CHECKING METHOD, TUBE ASSEMBLIES	39
FIGURE 19	DRIVEN BUTTON GAP CHECKING METHOD	40
FIGURE 20	FLUSH HEAD GAP CHECKING METHOD (EXCEPT BACR15FV AND NASM14218)	40
FIGURE 21	PART SEPARATION WITHOUT FILLER	41
FIGURE 22	PART SEPARATION WITH FILLER	42
FIGURE 23	PART SEPARATION IN JOGGLE	42

LIST OF TABLES

Table	Title	Page
TABLE I	HOLE SIZES (INCH) FOR SOLID RIVETS (EXCEPT NASM83459) AND RECOMMENDED TOOLING FOR MANUAL HOLE PREPARATION AND MEASUREMENT	9
TABLE II	HOLE SIZES (INCH) FOR NASM83459 RIVETS AND RECOMMENDED TOOLING FOR MANUAL HOLE PREPARATION AND MEASUREMENT	11
TABLE III	COUNTERSINK SIZES (INCH) FOR BACR15FV AND NASM14218 SOLID RIVET COUNTERSINKS PRODUCED BY ST1221V OR WST127535 COUNTERSINK CUTTERS MADE TO THE DIMENSIONS SPECIFIED IN Table III	12
TABLE IV	COUNTERSINK SIZES (INCH) FOR BACR15GF RIVET COUNTERSINKS PRODUCED BY ST1016-A-4EB, WST128049, OR ST1221JF 100° COUNTERSINK/ COUNTERBORE CUTTERS	13
TABLE V	COUNTERBORE/COUNTERSINK DEPTH (INCH) MEASUREMENT FOR BACR15FV AND NASM14218 SOLID RIVET	14
TABLE VI	COUNTERSINK DEPTH (INCH) MEASUREMENT FOR BACR15GF RIVET COUNTERSINKS PRODUCED BY ST1016-A-4EB, WST128049, OR ST1221JF 100° COUNTERSINK/ COUNTERBORE CUTTERS	15
TABLE VII	HOLE SIZE (INCH) FOR TUBULAR RIVETS	15
TABLE VIII	GRIP RANGE AND RECOMMENDED RIVET LENGTH DIMENSIONS (INCHES) FOR NASM83459 TITANIUM (CHERRYBUCK) RIVETS	17
TABLE IX	GRIP RANGE AND RECOMMENDED RIVET LENGTH FOR STANDARD ALUMINUM SOLID RIVETS	18

4 **CONTENTS (Continued)**

Table	Title	Page
TABLE X	DIMENSIONS (INCH) FOR SHAVING 100-DEGREE HEADS (EXCEPT BACR15CE, BACR15GF, BACR15FV, NASM14218 AND NAS1097)	20
TABLE XI	BACR15FV AND NASM14218 RIVET REPLACEMENT	24
TABLE XII	INSPECTION RELIABILITY REQUIREMENT (IRR)	27
TABLE XIII	DRIVEN RIVET BUTTON DIMENSIONS (INCH) FOR ALUMINUM RIVETS	30
TABLE XIV	DRIVEN RIVET BUTTON DIMENSIONS (INCH) FOR CRES AND NICKEL-COPPER ALLOY RIVETS (DOES NOT INCLUDE TUBULAR RIVETS)	32
TABLE XV	DRIVEN RIVET BUTTON DIMENSIONS (INCH) FOR NASM83459 TITANIUM (CHERRYBUCK) RIVETS	32
TABLE XVI	REQUIRED PROTRUSION FOR TUBULAR RIVETS	33
TABLE XVII	MINIMUM UNIVERSAL HEAD HEIGHT (INCH)	35
TABLE XVIII	MAXIMUM ALLOWABLE HEIGHT OF MARKS ON SINGLE IMPACT DRIVEN RIVET BUTTON	36

5 MATERIALS CONTROL

Not applicable to this specification.

6 FACILITIES CONTROL

6.1 CERTIFICATION OF EQUIPMENT

6.1.1 HYDRAULIC-SQUEEZE EQUIPMENT FOR BACR15FV, NAS1097, BACR15CE, AND BACR15GF RIVETS

a. All hydraulic-squeeze equipment used to install BACR15FV, NAS1097, BACR15CE, and BACR15GF countersunk rivets in fuselage skin structure (inclusive of wheel wells and areas covered by fairings), to the requirements of this specification, shall be certified in accordance with the requirements of D6-56617.

b. In the event that certified hydraulic-squeeze equipment is not available to produce parts, non-certified hydraulic-squeeze equipment may be used on a temporary basis provided:

(1) The process verification requirements of Section 8.2 are met.

(2) Statistical process control or in-process verification of rivet flushness, rivet hole diameter, and rivet driven button diameter and height is conducted in accordance with the requirements of Section 8.3 and this specification.

(3) Boeing approval is obtained for a temporary basis exceeding 90 days.

7 DEFINITIONS

The following definitions apply to terms that are uncommon or have special meaning as used in this specification.

Hydraulic Squeeze Equipment - A hydraulic-squeeze drill/rivet upset machine, such as a Drivmatic, Brotje, or Gemcor.

NACA Method - A flush riveting procedure introduced in a bulletin (Dec 1942) by the National Advisory Committee of Aeronautics (NACA). Generally, NACA driving is installing a rivet by driving to fill the countersink with the shank end (bucking the driven button into the countersink).

8 MANUFACTURING CONTROL

8.1 GENERAL

The following items apply to all solid rivets except as noted in the subsequent detail section or on applicable Engineering drawings.

8.1.1 RIVET INSTALLATION

a. Installation of BACR15FV and NASM14218 rivets, and BACR15GF rivets prepared using ST1016-A-4EB, WST128049, or ST1221JF countersink cutters, shall be accomplished through the use of a Process Control Document (PCD) to ensure proper control of the fastener installation process.

The PCD shall be approved by Manufacturing and Quality Assurance, and shall contain, as a minimum, information and process control data for establishing a reliable process for rivet installation, meeting the requirements of this specification. Instructions shall include information such as:

(1) Part fit-up

(2) Drill and cutter selection

(3) Lubricant selection

(4) Drill motor selection

(5) Speed and feed rates

(6) Hole diameter and hole alignment control

(7) Temporary protective coating thickness evaluation

(8) Countersink depth control

(9) Protection of clad skin

(10) Deburring operations

(11) Sealant removal from holes, rivets, bucking bars, and rivet gun dies

(12) Correct selection and alignment of rivet guns and bucking bars

(13) Verifying driven rivet button diameter and height

(14) Verifying requirements for no excessive gapping, driven rivet flushness, and no sharp edges or burrs

(15) Verifying flushness requirements

(16) Ensuring other specification and drawing unique requirements are met

b. For BACR15FV, NAS1097, BACR15CE, and BACR15GF countersunk rivets installed by hydraulic-squeeze equipment in fuselage skin structure, clamping pressure shall be adjusted to ensure that the workpiece components are firmly clamped together so that there is no gap at the hole location during hole preparation and rivet upset.

8.1.2 HOLES AND COUNTERSINKING

a. Hole sizes for protruding and flush head solid rivets, except titanium cherrybuck (NASM83459) rivets, shall conform to Table I.

8.1.2 HOLES AND COUNTERSINKING (Continued)

b. Hole sizes for titanium cherrybuck (NASM83459) rivets shall conform to Table II.

c. Hole sizes for tubular rivets shall conform to Table VII.

d. When dimpled holes are required, refer to BAC5049.

e. Exit burrs shall conform to the requirements of BAC5300. A hole exit burr of 0.005 inch maximum is permissible when the fastener is installed with automatic riveting equipment.

f. Countersinking depths for countersunk rivets shall be adjusted to meet flushness requirements, as applicable.

g. Counterbore/countersink depth for BACR15FV and NASM14218 rivets, prepared using ST1221V or WST127535 countersink cutters, shall not exceed Dimension B maximum, as shown in Table III and shall be verified in accordance with Table V using the WCI8703X or ST8703XB gage, or any suitable Quality Assurance certified variable data gage capable of meeting the requirements of Table V.

CAUTION Countersink depths for BACR15FV and NASM14218 rivets that exceed the B dimension in Table III may result in a knife-edge condition in thin skins.

h. Counterbore/countersink cutters for BACR15FV and NASM14218 rivets shall not be resharpened.

i. Countersinks for BACR15GF rivets, in clad aluminum fuselage aerodynamic surface only, may be prepared for improved appearance and paint adhesion using cutters which produce 100° countersinks with 20° included angle counterbore (100° CSK/CB). ST1016-A-4EB or WST128049 100° CSK/CB cutters may be used for automated applications and ST1221JF 100° CSK/CB cutters may be used for manual applications.

j. Countersink cutters ST1016-A-4EB, WST128049, and ST1221JF for BACR15GF rivets shall not be resharpened.

k. Countersink depth for BACR15GF rivets, prepared using ST1016-A-4EB, WST128049, or ST1221JF 100° CSK/CB cutters, shall not exceed Dimension B maximum as shown in Table IV, and shall be verified in accordance with Table VI using the WCI-8703CF gage or any suitable Quality Assurance certified variable data gage capable of meeting the requirements of Table VI.

l. Countersink depth, for BACR15FV rivets prepared using ST1221V or WST127535 countersink cutters, and for BACR15GF rivets installed with ST1016-A-4EB, WST128049, or ST1221JF 100° CSK/CB cutters, shall be controlled by In-Process Verification or Statistical Process Control (SPC), in accordance with Section 8.2 and Section 8.3. A documented verification plan (contained, for example, in a Process Control Document), approved by Manufacturing and Quality Assurance, is required if SPC is omitted. Re-verification is required if tooling (countersink cage, etc.) is adjusted or tooling wear is suspected.

8.1.2 HOLES AND COUNTERSINKING (Continued)

TABLE I - HOLE SIZES (INCH) FOR SOLID RIVETS (EXCEPT NASM83459) AND RECOMMENDED TOOLING FOR MANUAL HOLE PREPARATION AND MEASUREMENT

HOLE SIZE REQUIREMENTS FOR STANDARD RIVETS, INCLUDING BACR15AD, BACR15BA, BACR15BB, BACR15BC, BACR15CE, BACR15DS, BACR15ET, BACR15GF, BACR15FV, BACR15FT, NAS1097, NAS1198, NAS1199, NAS1200, NAS1242, NASM14218, NASM20426, NASM20427, NASM20470, NASM20613, NASM20615, (EXCEPT NASM83459, SEE Table II)				STANDARD TOOL DRAWINGS AND DETAILS					
				DRILL FL 1 FL 2	REAMER FL 1 FL 2	COUNTERSINK CUTTER			GO/NO-GO GAGE
				ST10-937-B OR ST7101AB FL 3 FL 4	ST1864L FL 3	ST1221JD-A FL 3 FL 5 FL 6 FL 7	ST1221JF FL 5 FL 6 FL 8	ST1221V FL 6 FL 9	ST8725D FL 10
RIVET DIAMETER CODE FL 11	NOMINAL RIVET DIAMETER	HOLE DIAMETER LIMITS		DIAMETER (DRILL SIZE)	MAJOR DIA.- PILOT DIA.	100° CSK. (STANDARD)	100° CSK/ CB	120° CSK/ CB	GAGE DIAMETER CODE
		MIN	MAX			PILOT DIAMETER OR DETAIL CODE			
2	1/16 (.0625)	0.066	0.072	0.067 (51)	- - -	-065	- - -	- - -	- - -
3	3/32 (.0938)	0.098	0.103	0.098 (40)	- - -	-096	- - -	-3	- - -
4	1/8 (.1250)	0.128	0.134	0.1285 (30)	- - -	-126	- - -	-4	- - -
5	5/32 (.1563)	0.159	0.165	0.161 (20) 0.1600	- - -	-158	-5	-5B	-5RD
6	3/16 (.1875)	0.190	0.196	0.1920	- - -	-190	-6	-6B	-6RD
61	13/64 (.2031)	0.205	0.209	0.2050	2060-191	-203	- - -	-61	- - -
7	7/32 (.2188)	0.225	0.231	0.2260	2260-191 2260-204	-224	-7	-7B	-7RD
71	15/64 (.2344)	0.239	0.243	0.2390	2400-224	-238	- - -	-71	- - -
8	1/4 (.2500)	0.253	0.259	0.2540	2540-217 2540-225 2540-238	-252	-8	-8B	-8RD
81	17/64 (.2656)	0.268	0.272	0.2680	2690-253	-266	- - -	-81	- - -
9	9/32 (.2813)	0.286	0.292	0.2870	2870-249 2870-253 2870-267	-285	- - -	-9B	-9RD
91	19/64 (.2969)	0.300	0.304	0.3010	3010-286	- - -	- - -	-91	- - -
10	5/16 (.3125)	0.316	0.322	0.3160	3160-280 3160-300	-314	- - -	-10B	-10R
11	11/32 (.3437)	0.348	0.358	0.348 (S)	3490-311 3490-315	-346	- - -	- - -	- - -
12	3/8 (.3750)	0.377	0.385	0.377 (V)	3780-342	-375	- - -	- - -	- - -
13	13/32 (.4063)	0.407	0.421	0.413 (Z)	4130-376	-411	- - -	- - -	- - -

8.1.2 HOLES AND COUNTERSINKING (Continued)

FL 1 The target hole diameter shall be as small as practical. The target hole diameter shall fall within the specification limits. The target hole diameter shall allow free insertion of the rivet. Scraping of the rivet or the hole is not acceptable.

FL 2 Recommended reamer major diameters and reamer pilot diameters are sized to allow clearance between the countersink pilot or previous step's cutter. Reamer major diameters are sized .002 to .003 larger than the countersink pilot diameters.

Example: Countersink pilot diameter for 11/32 is .346; Reamer major diameter is .3490.

Multiple pilot sizes for reamers listed are based on the previous operation. When the previous operation was creation of a pilot hole produced with a fractional drill, the reamer pilot diameter is .001 smaller than the fractional drill diameter.

Example: Following a 1/4 (.250) drill, use a .249 reamer pilot diameter.

When the previous operation requires fastener replacement with a larger size, the reamer pilot diameters are sized .001 to .002 smaller than the previous step's recommended cutter diameter. Example: Previous recommended cutter was .254; recommended pilot is .253.

FL 3 National Aerospace Standard (NAS) or Boeing Tool Commodity Standard (BTCS) tool drawings are acceptable substitutions.

FL 4 Drill sizes are identified by number, letter, or major diameter. Stepped, double margin drills are recommended for diameters between 5/32 and 5/16.

FL 5 Use ST1221JD-A countersink cutters for BACR15GF 61, 81 and 10 diameter codes.

FL 6 Countersink pilot diameters sized .002 to .004 less than the recommended cutter diameter.

FL 7 No specific gage is recommended to measure standard 100° countersink dimensions. Use of countersink diameter or depth gage is acceptable to control the process. Countersink depth and upset force are adjusted to achieve the required installation flushness.

FL 8 Use WCI8703CF or equivalent gage to measure countersink depth of 100° countersink with 20° included angle counterbore (100° CSK/CB) to meet requirements of Table IV.

FL 9 Use ST8703XB, or WCI8703X or equivalent certified gage to measure countersink depth to meet requirements of Table III.

FL 10 Use ST8725D go/no-go diameter gage, available as indicated in table. See ST1810D-KB and BSS7022 for other measurement options.

FL 11 Rivet diameter codes 61, 7, 71, 81, 9, 91, 11 and 13 are oversize. Rivet diameter codes 7 and 9 may be also used as nominal; see Engineering drawing for applicability.

8.1.2 HOLES AND COUNTERSINKING (Continued)

TABLE II - HOLE SIZES (INCH) FOR NASM83459 RIVETS AND RECOMMENDED TOOLING FOR MANUAL HOLE PREPARATION AND MEASUREMENT

HOLE SIZE REQUIREMENTS FOR NASM83459				STANDARD TOOL DRAWINGS AND DETAILS			
				DRILL FL 1	REAMER FL 1	COUNTERSINK CUTTER	GO/NO-GO GAGE
				ST10-937-B OR ST7101AB FL 2 FL 3	ST1864L FL 2	ST1221JD FL 4 FL 5	ST88725D FL 6
RIVET DIAMETER CODE	NOMINAL RIVET DIAMETER	HOLE DIAMETER LIMITS		DIAMETER (DRILL SIZE)	MAJOR DIA.-PILOT DIA.	100° CSK.	GAGE DIAMETER CODE
		MIN	MAX			PILOT DIAMETER	
5	5/32 (.1563)	0.1600	0.1630	0.1600	1600-124	-158	MISC1600-MISC1630
6	3/16 (.1875)	0.1855	0.1885	0.1885	1855-155	-184	MISC1855-MISC1885
8	1/4 (.2500)	0.2455	0.2485	0.2455	2455-217	-244	MISC2455-MISC2485
10	5/16 (.3125)	0.3080	0.3110	0.3080	3080-280	-306	MISC3080-MISC3110
12	3/8 (.3750)	0.3705	0.3735	- - -	3705-342	-368	MISC3705-MISC3735

FL 1 Reaming is recommended for diameters 3/8 or greater. Reamer pilot diameters are sized .001 below the fractional pilot hole size.

FL 2 National Aerospace Standard (NAS) or Boeing Tool Commodity Standard (BTCS) tool drawings are acceptable substitutions.

FL 3 Drill sizes are identified by major diameter. Stepped, double margin drills are recommended for diameters between 5/32 and 5/16.

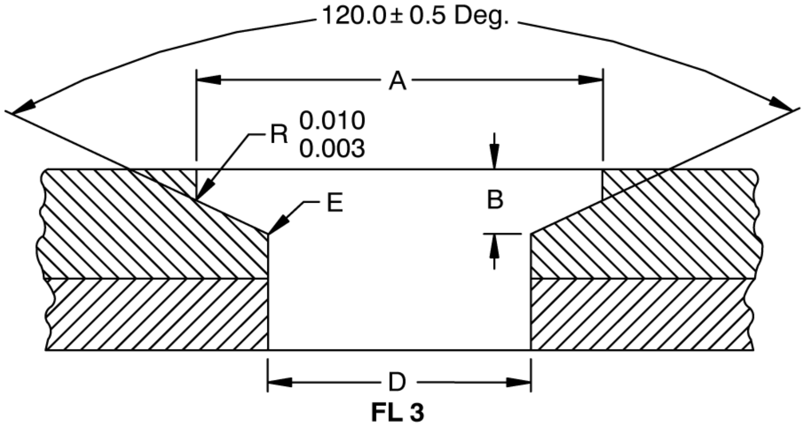
FL 4 Countersink pilot diameters sized .002 to .004 less than the recommended cutter diameter.

FL 5 No specific gage is recommended to measure standard 100° countersink dimensions. Use of countersink diameter or depth gage is acceptable to control the process.

FL 6 Use ST8725D go/no-go diameter gage MISC option. See ST1810D-KB and BSS7022 for other measurement options.

8.1.2 HOLES AND COUNTERSINKING (Continued)

TABLE III - COUNTERSINK SIZES (INCH) FOR BACR15FV AND NASM14218 SOLID RIVET COUNTERSINKS PRODUCED BY ST1221V OR WST127535 COUNTERSINK CUTTERS MADE TO THE DIMENSIONS SPECIFIED IN Table III



RIVET DIAMETER CODE	NOMINAL RIVET DIAMETER	A		B MAXIMUM FL 1 FL 2	E RAD ± 0.004
		MIN	MAX		
4	1/8	0.1625	0.1655	0.028	0.008
5	5/32	0.2105	0.2140	0.035	0.009
6	3/16	0.2585	0.2625	0.044	0.012
61	13/64	0.2735	0.2780	0.044	0.013
7	7/32	0.3125	0.3170	0.053	0.015
71	15/64	0.3265	0.3310	0.053	0.016
8	1/4	0.3585	0.3640	0.061	0.018
81	17/64	0.3735	0.3790	0.061	0.019
9	9/32	0.4055	0.4110	0.069	0.020
91	19/64	0.4195	0.4250	0.069	0.021
10	5/16	0.4545	0.4600	0.078	0.022

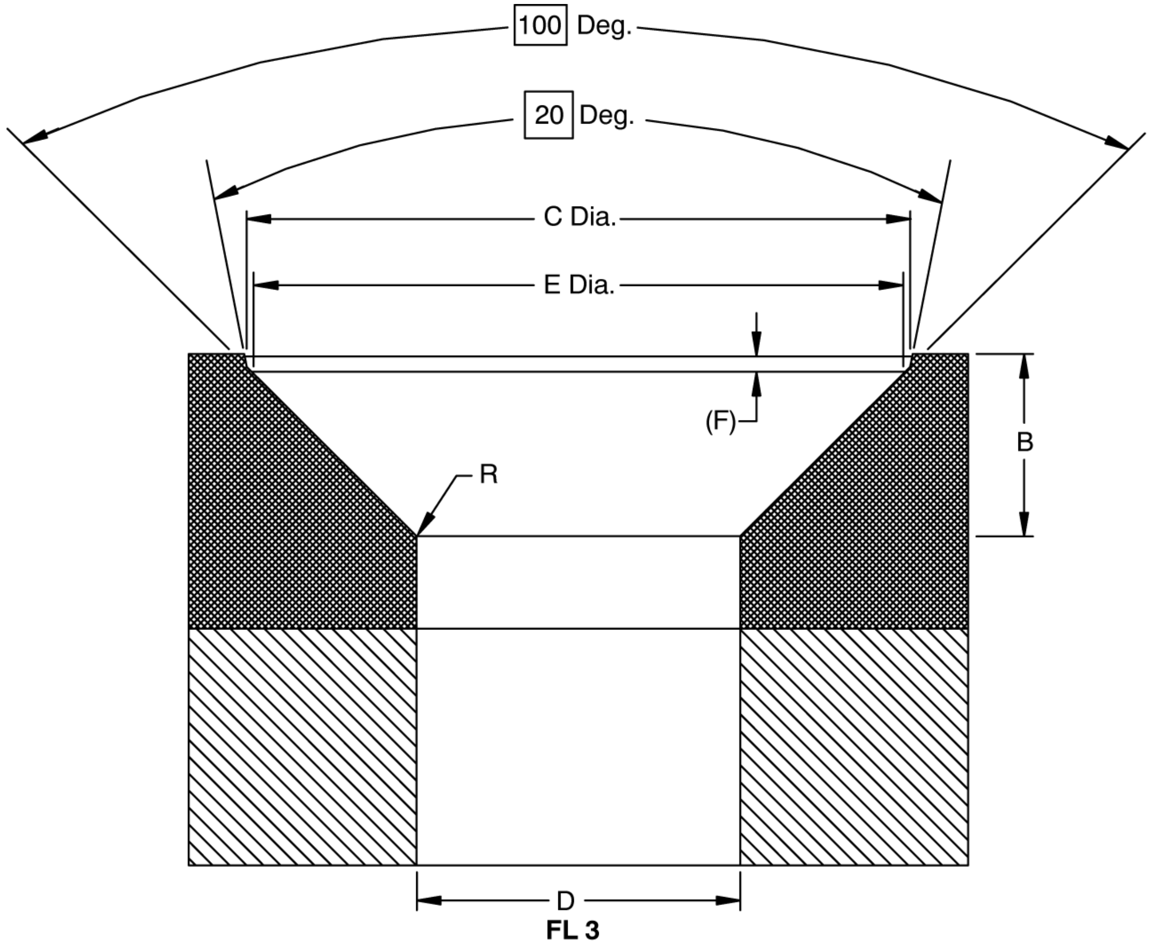
FL 1 B dimension is the distance from the top of the skin to the theoretical intersection of the countersink and the hole at the minimum hole diameter.

FL 2 Table V shall be used for counterbore/countersink measurement.

FL 3 Hole diameter (D dimension) for BACR15FV and NASM14218 rivets is shown in Table I.

8.1.2 HOLES AND COUNTERSINKING (Continued)

TABLE IV - COUNTERSINK SIZES (INCH) FOR BACR15GF RIVET COUNTERSINKS PRODUCED BY ST1016-A-4EB, WST128049, OR ST1221JF 100° COUNTERSINK/COUNTERBORE CUTTERS



RIVET DIAMETER CODE	NOMINAL RIVET DIAMETER	B FL 1 FL 2		C DIAMETER		E DIAMETER		F REF		R RADIUS	
		MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.
5	5/32	0.0359	0.0314	0.2388	0.2364	0.2379	0.2359	0.0036	0.0008	0.010	0.005
6	3/16	0.0439	0.0394	0.2886	0.2862	0.2876	0.2856	0.0038	0.0009	0.015	0.005
7	7/32	0.0438	0.0393	0.3227	0.3203	0.3220	0.3200	0.0031	0.0004	0.015	0.005
8	1/4	0.0578	0.0533	0.3878	0.3854	0.3870	0.3850	0.0032	0.0004	0.015	0.005

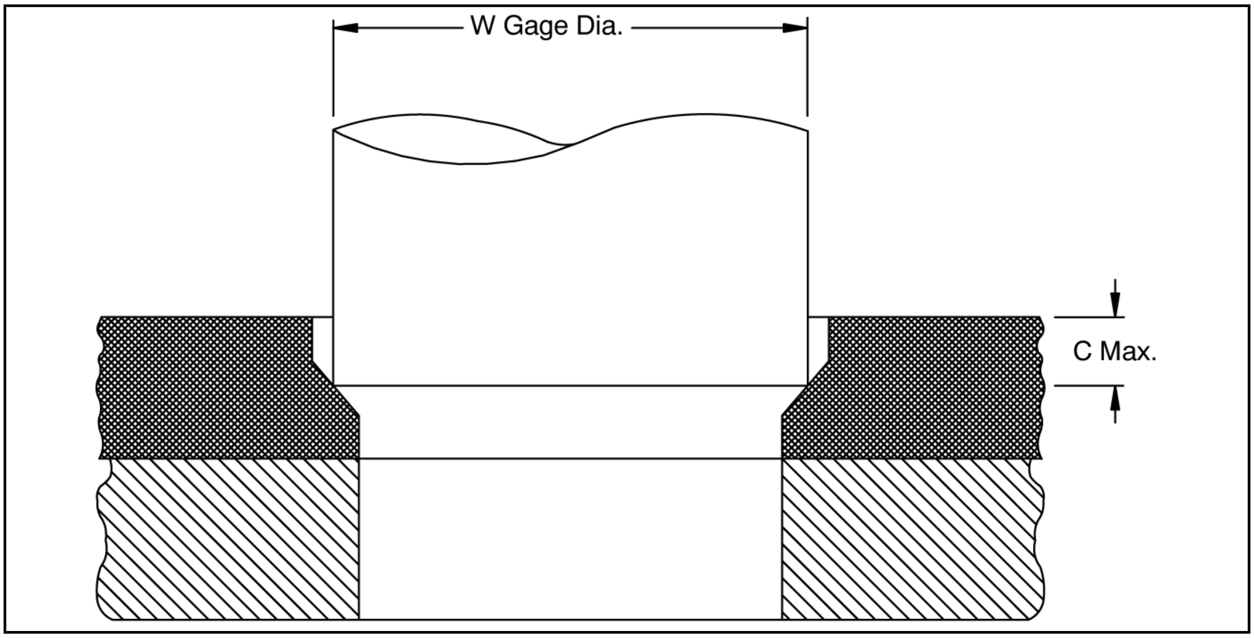
FL 1 B dimension is the distance from the top of the skin to the theoretical intersection of the countersink and the hole at the minimum hole diameter.

FL 2 Table VI shall be used for countersink measurement.

FL 3 Hole diameter (D dimension) for BACR15GF rivets prepared using ST1016-A-4EB, WST128049, or ST1221JF 100° CSK/CB cutters is identical to D dimensions for BACR15GF rivets in Table I.

8.1.2 HOLES AND COUNTERSINKING (Continued)

TABLE V - COUNTERBORE/COUNTERSINK DEPTH (INCH) MEASUREMENT FOR BACR15FV AND NASM14218 SOLID RIVET

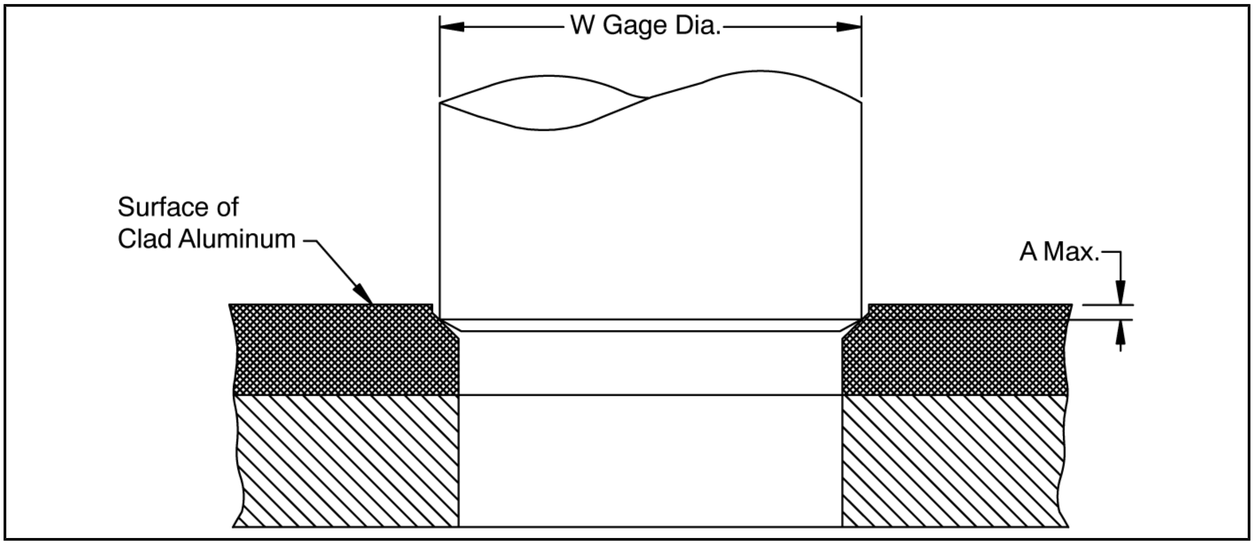


RIVET DIAMETER CODE	NOMINAL RIVET DIAMETER	W GAGE DIAMETER		C MAX FL 1
		MINIMUM	MAXIMUM	
4	1/8	0.1523	0.1525	0.0204
5	5/32	0.2003	0.2005	0.0229
6	3/16	0.2483	0.2485	0.0269
61	13/64	0.2483	0.2485	0.0314
7	7/32	0.3023	0.3025	0.0313
71	15/64	0.3023	0.3025	0.0345
8	1/4	0.3483	0.3485	0.0337
81	17/64	0.3483	0.3485	0.0376
9	9/32	0.3953	0.3955	0.0382
91	19/64	0.3953	0.3955	0.0412
10	5/16	0.4243	0.4245	0.0464

FL 1 For each diameter, C max corresponds to B maximum (at minimum hole diameter) found in Table III.

8.1.2 HOLES AND COUNTERSINKING (Continued)

TABLE VI - COUNTERSINK DEPTH (INCH) MEASUREMENT FOR BACR15GF RIVET COUNTERSINKS PRODUCED BY ST1016-A-4EB, WST128049, OR ST1221JF 100° COUNTERSINK/COUNTERBORE CUTTERS



RIVET DIAMETER CODE	NOMINAL RIVET DIAMETER	W GAGE DIAMETER		A MAX FL 1 FL 2 FL 3
		MINIMUM	MAXIMUM	
5	5/32	0.2109	0.2111	0.0139
6	3/16	0.2438	0.2440	0.0211
7	7/32	0.2438	0.2440	0.0349
8	1/4	0.3312	0.3314	0.0256

FL 1 Dimension A is the distance between the clad aluminum surface (without temporary protective coating) and the point of contract of the gage with the countersink.

FL 2 For each diameter, A max corresponds to B max (at minimum hole diameter) found in Table IV.

FL 3 For ST1221JF manual cutter only, A max dimension may be exceeded by up to 0.0020 to obtain required flushness.

TABLE VII - HOLE SIZE (INCH) FOR TUBULAR RIVETS

NOMINAL RIVET DIAMETER	RECOMMENDED PILOT DRILL SIZE	HOLE DIAMETER LIMITS	
		MINIMUM	MAXIMUM
3/32	#39	0.098	0.103
1/8	1/8	0.125	0.129
5/32	#26	0.147	0.151
3/16	12	0.188	0.192

8.1.3 HOLE LOCATION

Rivet manufactured heads and driven rivet buttons may be located as indicated in Figure 1.

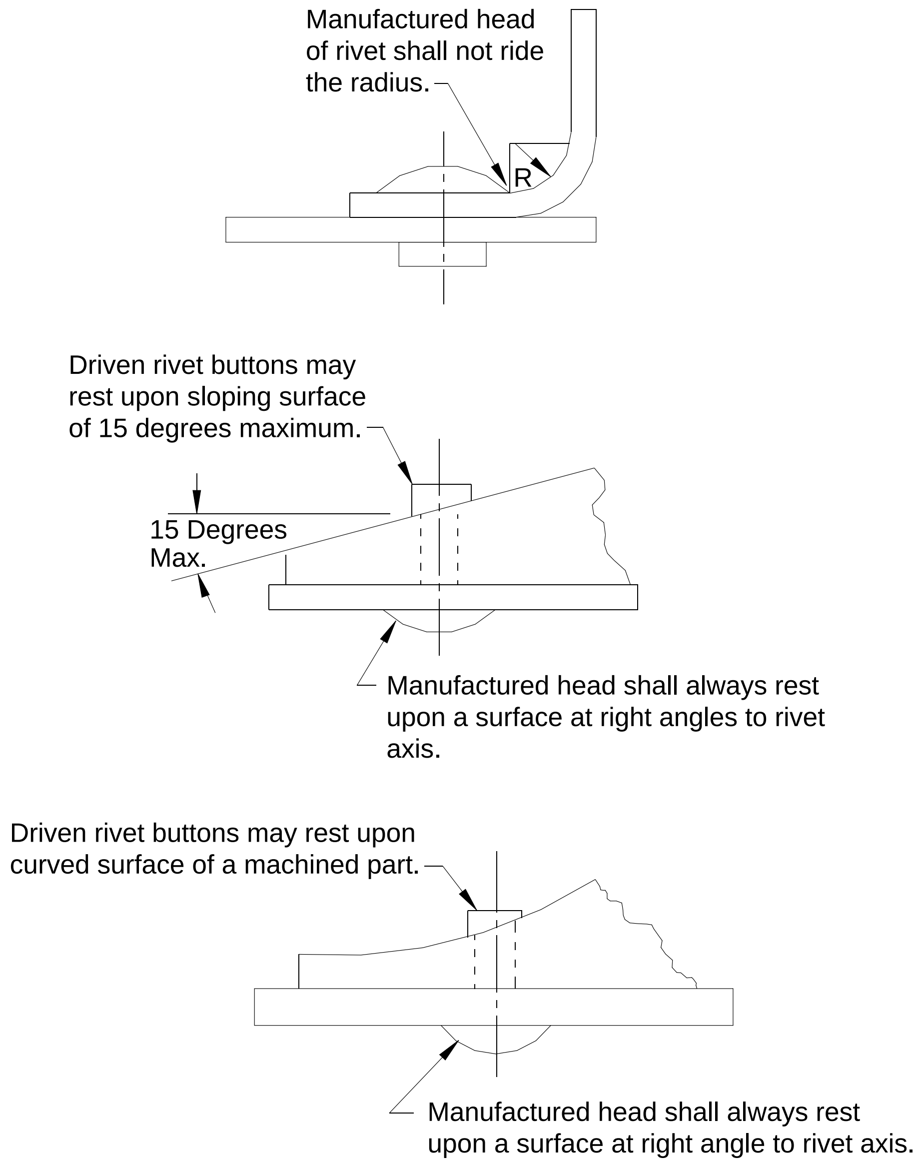


FIGURE 1 - RIVET HOLE LOCATION

8.1.4 RIVET LENGTH

a. Use Figure 2 and Table VIII as a guide in determining the proper rivet length dash number for NASM83459 titanium (Cherrybuck) rivets. NASM83459 (Cherrybuck) rivet lengths shall be selected to meet the driven button dimensions shown in Table XV.

8.1.4 RIVET LENGTH (Continued)

b. Use Table IX and Figure 3 as a guide in determining the proper rivet length dash number for standard aluminum solid rivets. Standard rivet lengths shall be selected to meet the driven button dimensions shown in Table XIII.

c. Use Table IX and Figure 3 also as a guide in determining the proper rivet length dash number for CRES and nickel-copper solid rivets. Rivet lengths shall be selected to meet the driven button dimensions shown in Table XIV.

d. Use Table XVI as a guide in determining the proper rivet length dash number for tubular rivets.

e. Adjustment of drawing-specified rivet length is permitted when necessary. Driven button dimensions for the selected length, in accordance with Section 8.1.7, and all other requirements specified herein shall apply.

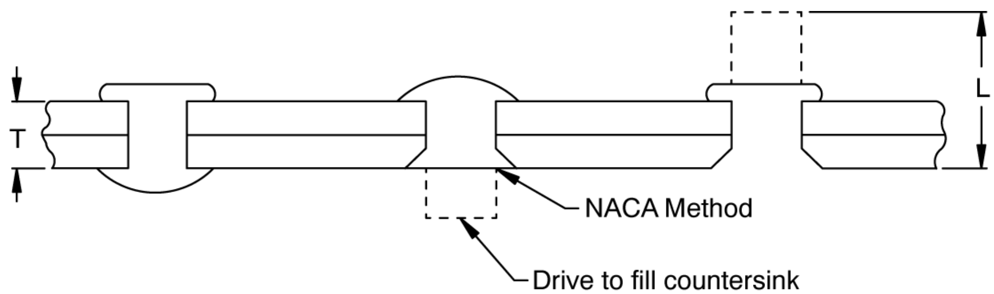


FIGURE 2 - RIVET LENGTH (L) AND GRIP (T) DIMENSIONS FOR NASM83459 (CHERRYBUCK) RIVETS

TABLE VIII - GRIP RANGE AND RECOMMENDED RIVET LENGTH DIMENSIONS (INCHES) FOR NASM83459 TITANIUM (CHERRYBUCK) RIVETS FL 1 FL 2

T GRIP RANGE		5/32 (-5) DIA.		3/16 (-6) DIA.		1/4 (-8) DIA.		5/16 (-10) DIA.		3/8 (-12) DIA.	
		RIVET LENGTH DASH NO. AND DIMENSIONS									
MIN	MAX.	DASH NO.	L ±0.010	DASH NO.	L ±0.010	DASH NO.	L ±0.010	DASH NO.	L ±0.010	DASH NO.	L ±0.010
0.094	0.125	5-2	0.298	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -
0.126	0.187	5-3	0.360	6-3	0.385	- - -	- - -	- - -	- - -	- - -	- - -
0.188	0.250	5-4	0.422	6-4	0.447	8-4	0.502	10-4	0.577	- - -	- - -
0.251	0.312	5-5	0.485	6-5	0.510	8-5	0.565	10-5	0.620	12-5	0.675
0.313	0.375	5-6	0.547	6-6	0.572	8-6	0.627	10-6	0.682	12-6	0.737
0.376	0.437	5-7	0.610	6-7	0.635	8-7	0.690	10-7	0.745	12-7	0.800
0.438	0.500	5-8	0.672	6-8	0.697	8-8	0.752	10-8	0.807	12-8	0.862
0.501	0.562	5-9	0.735	6-9	0.760	8-9	0.815	10-9	0.870	12-9	0.925
0.563	0.625	5-10	0.797	6-10	0.822	8-10	0.877	10-10	0.932	12-10	0.987
0.626	0.687	5-11	0.860	6-11	0.885	8-11	0.940	10-11	0.995	- - -	- - -
0.688	0.750	5-12	0.922	6-12	0.947	8-12	1.002	10-12	1.057	- - -	- - -
0.751	0.812	5-13	0.985	6-13	1.010	8-13	1.065	10-13	1.120	- - -	- - -
0.813	0.875	5-14	1.047	6-14	1.072	8-14	1.127	10-14	1.182	- - -	- - -

8.1.4 RIVET LENGTH (Continued)

TABLE VIII - GRIP RANGE AND RECOMMENDED RIVET LENGTH DIMENSIONS (INCHES) FOR NASM83459 TITANIUM (CHERRYBUCK) RIVETS FL 1 FL 2 (Continued)

0.876	0.927	5-15	1.110	6-15	1.135	8-15	1.190	10-15	1.245	- - -	- - -
0.938	1.000	5-16	1.172	6-16	1.197	8-16	1.252	10-16	1.307	- - -	- - -

FL 1 Do not cut rivet shank end to obtain shorter rivet length.

FL 2 Do not use in less than specified minimum grip.

TABLE IX - GRIP RANGE AND RECOMMENDED RIVET LENGTH FOR STANDARD ALUMINUM SOLID RIVETS

RIVET LENGTH DASH NUMBER	RECOMMENDED GRIP RANGE IN INCHES FL 1 FL 2 FL 3 FL 4								
	3/32 OR 3 DIA-METER	1/8 OR 4 DIA-METER	5/32 OR 5 DIA-METER	3/16 OR 6 DIA-METER	7/32 OR 7 DIA-METER	1/4 OR 8 DIA-METER	9/32 OR 9 DIA-METER	5/16 OR 10 DIA-METER	3/8 OR 12 DIA-METER
-3	0.084	-0.058	-0.035	- - -	- - -	- - -	- - -	- - -	- - -
-4	0.085-0.142	0.059-0.117	0.036-0.093	-0.068	- - -	- - -	- - -	- - -	- - -
-5	0.143-0.201	0.118-0.176	0.094-0.152	0.069-0.127	-0.107	-0.078	-0.058	- - -	- - -
-6	0.202-0.260	0.177-0.234	0.153 -0.211	0.128-0.185	0.108-0.166	0.079-0.136	0.059-0.117	-0.087	-0.038
-7	0.261-0.319	0.235-0.293	0.212-0.270	0.186-0.244	0.167-0.225	0.137-0.195	0.118-0.176	0.088-0.146	0.039-0.097
-8	0.320-0.377	0.294-0.352	0.271-0.327	0.245-0.303	0.226-0.283	0.196-0.254	0.177-0.234	0.146-0.205	0.098-0.156
-9	0.378-0.436	0.353-0.411	0.328-0.387	0.304-0.362	0.284-0.342	0.255-0.313	0.235-0.293	0.206-0.264	0.157-0.215
-10	0.437-0.495	0.412-0.469	0.388-0.446	0.363-0.420	0.343-0.401	0.314-0.371	0.294-0.352	0.265-0.322	0.216-0.273
-11	- - -	0.470-0.528	0.447-0.505	0.421-0.479	0.402-0.460	0.372-0.430	0.353-0.411	0.323-0.381	0.274-0.332
-12	- - -	0.529-0.587	0.506-0.563	0.480-0.538	0.461-0.518	0.431-0.489	0.412-0.469	0.382-0.440	0.333-0.391
-13	- - -	0.588-0.646	0.564-0.622	0.539-0.597	0.519-0.577	0.490-0.548	0.470-0.528	0.441-0.499	0.392-0.450
-14	- - -	- - -	0.623-0.681	0.598-0.655	0.578-0.636	0.549-0.606	0.529-0.587	0.500-0.557	0.451-0.508
-15	- - -	- - -	0.682-0.740	0.656-0.714	0.634-0.695	0.607-0.665	0.588-0.646	0.558-0.616	0.509-0.567
-16	- - -	- - -	0.741-0.798	0.715-0.773	0.696-0.753	0.666-0.724	0.647-0.704	0.617-0.675	0.568-0.626
-17	- - -	- - -	- - -	0.774-0.832	0.754-0.812	0.725-0.783	0.705-0.763	0.676-0.734	0.627-0.685

8.1.4 RIVET LENGTH (Continued)

TABLE IX - GRIP RANGE AND RECOMMENDED RIVET LENGTH FOR STANDARD ALUMINUM SOLID RIVETS (Continued)

-18	- - -	- - -	- - -	0.833-0.890	0.813-0.871	0.784-0.841	0.764-0.822	0.735-0.792	0.686-0.743
-19	- - -	- - -	- - -	0.891-0.949	0.872-0.930	0.842-0.900	0.823-0.881	0.793-0.851	0.744-0.802
-20	- - -	- - -	- - -	0.950-1.008	0.931-0.988	0.901-0.959	0.882-0.939	0.852-0.910	0.803-0.861
-21	- - -	- - -	- - -	- - -	0.989-1.047	0.960-1.018	0.940-0.998	0.911-0.969	0.862-0.920
-22	- - -	- - -	- - -	- - -	- - -	1.019-1.076	0.999-1.057	0.970-1.027	0.921-0.978

FL 1 The dash number represents the nominal length in 1/16ths of an inch.

FL 2 For rivets requiring 1.4X diameter driven button, the next half-sized (R5 suffix to part number) length may be required. For rivets requiring 1.5X diameter driven button, the next whole dash number may be required.

FL 3 In thick stackups, for rivets installed in holes that are the maximum allowed diameter, the next half-sized (R5) length may be required.

FL 4 In minimum stack-ups the previous half sized (R5 suffix to part number) length may be used for weight savings.

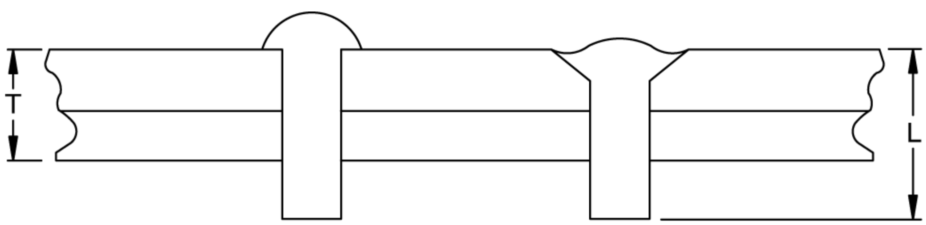


FIGURE 3 - RIVET LENGTH (L) AND GRIP (T) DIMENSIONS FOR STANDARD ALUMINUM SOLID RIVETS

8.1.5 ALLOYS AND HEAT TREATMENT

The 2024 (DD) rivets shall be heat treated and stored in accordance with BAC5602 and driven within 15 minutes of removal from cold storage.

NOTE: The purpose of the cold storage is to retard the age hardening which occurs at temperatures above -10 F. Once age hardening has begun, returning them to cold storage will not soften them.

8.1.6 FLUSH RIVETING

a. When shaving the manufactured head of solid rivets, including rivets in tanks and pressurized sections, surface flushness shall be in accordance with Table X and Section 11.1.1.

(1) Head protrusion shall not exceed H in Table X after driving and prior to shaving.

(2) Head diameter shall not be less than D in Table X after shaving.

(3) BACR15CE, BACR15GF, BACR15FV, NASM14218 and NAS1097 rivets shall not be shaved or modified (for example, sanding, grinding) to meet flushness requirements without Design Engineering approval, except as noted in Section 8.1.9.

b. Manufacturing tolerances on the heads of flush-head fasteners may result in partially exposed countersinks (see Figure 4). Exposed areas around the fastener head are acceptable if the fastener tolerances, hole tolerances, and flushness requirements are met. (Does not apply to BACR15FV and NASM14218 rivets).

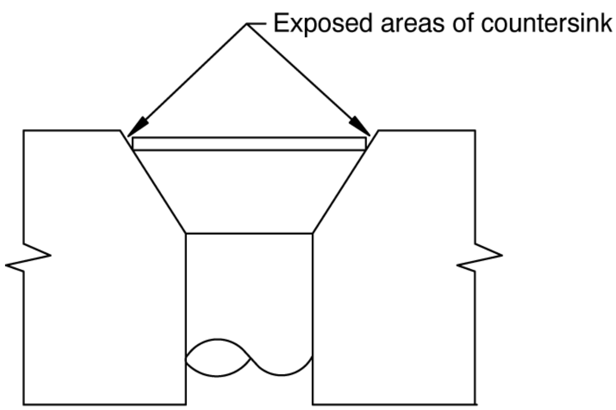
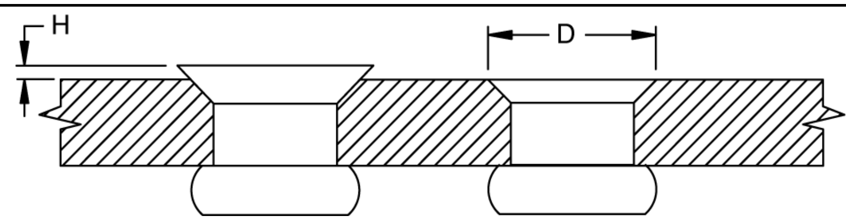


FIGURE 4 - PARTIALLY EXPOSED COUNTERSINK

TABLE X - DIMENSIONS (INCH) FOR SHAVING 100-DEGREE HEADS (EXCEPT BACR15CE, BACR15GF, BACR15FV, NASM14218 AND NAS1097)



NOMINAL RIVET DIAMETER (INCH)	3/32	1/8	5/32	3/16	1/4	5/16	3/8
H Maximum Protrusion Prior to Shaving	0.006	0.007	0.009	0.010	0.012	0.014	0.016
D Minimum Head Diameter After Shaving	0.161	0.204	0.262	0.326	0.443	0.526	0.650

8.1.7 DRIVEN BUTTONS

a. Rivets shall be driven to the driven button dimensions listed in Table XIII, Table XIV, or Table XV as applicable unless otherwise specified on the drawing.

b. Except for rivets installed by automatic riveting machines, aluminum (except 2024-T4) and NASM83459 titanium rivets may be re-hit as required.

c. When a flush head rivet, except BACR15FV and NASM14218, is driven double flush, the countersink for the driven button shall be the same as used for the manufactured head. The minimum driven button diameter is 1.5d as noted in Table XIII.

d. The flush driven button of BACR15FV and NASM14218 rivets may be driven into a BACR15FV, NASM14218, NAS1097, BACR15CE, or BACR15GF countersink. When a BACR15FV or NASM14218 countersink is used, gaps around driven buttons are not acceptable. When an NAS1097, BACR15CE or BACR15GF countersink is used, the minimum driven button diameter is 1.5d.

e. When a protruding head rivet is installed with the driven button flush (NACA method), the minimum button diameter is 1.5d. Use the drawing specified countersink.

f. All rivet buttons driven flush shall be shaved within +0.003, -0.000 inch.

8.1.8 TEMPORARY RIVET RETENTION

a. Universal (protruding) head rivets may be temporarily retained in place in restricted clearance areas of the body and empennage (such as frame-to-stringer clips) using adhesive applied in accordance with BAC5010 Type 48. Rivets shall be driven within 30 minutes of adhesive application. Excess adhesive shall be removed within 60 minutes after adhesive application in accordance with BAC5750. Adhesive shall not be applied to rivets that require under-head sealing.

b. Special rivet tapes in accordance with BAC5034, shall be used when holding inserted rivets in place for riveting.

CAUTION Because tape can become entrapped in the head dimple of BACR15FV and NASM14218 rivets during driving, giving the rivet the appearance of not having been driven, the use of only one layer of the correct tape is important to minimize this effect.

c. The presence of entrapped rivet tape under the rivet heads or in the countersinks of installed rivets is prohibited.

8.1.9 REWORK

The following rework shall be documented as required by the applicable quality assurance provisions. All other rework shall be authorized and documented as required by the applicable quality assurance provisions.

8.1.9 REWORK (Continued)

a. Except for BACR15FV, NASM14218 and NASM83459 rivets, installed solid rivets in sizes 1/16 inch diameter through 3/8 inch diameter may be replaced with 1/32 inch diameter larger rivets to accomplish rework provided the replacement rivet meets the centerline tolerance of the original rivet. Engineering approval and Quality Assurance verification are required if edge margin is less than two diameters; whereas only Quality Assurance verification is required if the edge margin is twice the diameter of the original fastener or greater.

Exception: Engineering approval and QA verification are required for 1/32 oversize rework of 3/32, 1/8, and 5/32 inch diameter countersunk solid rivet installations where the rework countersink depth is greater than 2/3 of the sheet thickness. An increase in countersink depth to oversize these diameters may result in an unacceptable knife-edge condition in structure.

b. Hole sizes for rivets 1/32 inch diameter larger than the original rivet callout are shown in Table I.

c. Quality Assurance verification is required for use of BACR15GF61, BACR15GF81, BACR15FV61, BACR15FV71, BACR15FV81, and BACR15FV91 1/64 oversize rivets, which have the same countersink depth requirement as the original size rivet. Engineering approval is also required if edge margin is less than two diameters. Hole sizes for (61, 71, 81, and 91 diameter code) rivets 1/64 inch diameter larger than the original rivet callout are shown in Table I.

(1) Countersinks for BACR15GF61 and BACR15GF81 1/64 oversize rivets shall be prepared using 100 degree standard countersink cutters. ST1016-A-4EB, WST128049, or ST1221JF 100° CSK/CB cutters shall not be used.

CAUTION Countersinks from previously installed rivets may be too deep to successfully install 1/64 oversize rivets and meet the installed flushness (minimum +0.001 inch) requirement.

(2) Holes that have been previously prepared using 100° CSK/CB cutters shall be touched up using 100 degree standard countersink cutters. The counterbore feature produced by 100° CSK/CB cutters shall be completely removed.

CAUTION Countersinks from previously installed rivets may have been prepared using 100° CSK/CB cutters.

(3) Driven button dimensions for BACR15GF61 and BACR15GF81 1/64 oversize rivets shall conform to Table XIII.

(4) Counterbores/countersinks for BACR15FV61, BACR15FV71, BACR15FV81, and BACR15FV91 1/64 oversize rivets shall be prepared using 1/64 oversize ST1221V cutters.

(5) Counterbore/countersink depth for BACR15FV61, BACR15FV71, BACR15FC81, and BACR15FV91 1/64 oversize rivets shall be verified in accordance with Table V using the ST8703XB gage or any suitable Quality Assurance certified variable data gage capable of meeting the requirements of Table V.

8.1.9 REWORK (Continued)

(6) Driven buttons for BACR15FV61, BACR15FV71, BACR15FC81, and BACR15FV91 1/64 oversize rivets shall be verified in accordance with Table XIII.

d. Installed BACR15FV and NASM14218 rivets may be replaced with 1/32 inch diameter larger rivets without Engineering approval (Quality Assurance verification required) provided:

(1) The replacement rivet meets the centerline tolerance of the original rivet.

(2) Edge margin is twice the diameter of the original fastener, or greater.

(3) The replacement rivet is installed in skin with allowable thickness (where the next larger diameter rivet still meets the requirements of Table XI). See Figure 5.

Install the next larger rivet in accordance with Section 8.1.2 and Section 11.1.1 or the appropriate Engineering drawing. Do not shave the head.

Engineering approval is required when the replacement rivet is installed, with full countersink depth, in skin with less than allowable thickness (where the next larger diameter rivet does not meet the requirement of Table XI.)

e. BACR15FV and NASM14218 rivets installed in thin skin (where the next 1/32 larger diameter rivet does not meet the requirements of Table XI) may be replaced with the next larger diameter rivet as follows, without Engineering approval (Quality Assurance verification is required), provided the countersink depth for the original size (1/8, 5/32, 3/16, 7/32, 1/4, 9/32, and 5/16) rivet is not exceeded. See Figure 6.

Increase the countersink diameter of the hole using the next size larger countersink cutter, except do not increase the original countersink depth as shown in Table III for the original size rivet. See Figure 6.

Ensure that counterbore/countersink depth shall not exceed that shown in Table III, Dimension B maximum for the original rivet. Verify using the WCI8703X or ST8703XB or equivalent gage for the next 1/32 larger diameter rivet.

The countersink depth measurement shall not be less than the difference in Bmax for the original rivet and Bmax for the next 1/32 larger diameter rivet.

Example:

To verify countersink depth for a 3/16 original rivet replaced with the next 1/32 larger diameter 7/32 rivet (where the original countersink depth is not exceeded), use the 7/32 gage (Cmax = .0223). The measurement shall not be less than .009 inch [.053 (7/32 Bmax) - .044 (3/16 Bmax) = .009].

Place the next 1/32 larger diameter rivet in the hole, and drive the rivet.

Microshave the head to meet the head height requirements of Section 11.1.1 or the applicable Engineering drawing.

8.1.9 REWORK (Continued)

f. Aluminum (except 2024-T4) and NASM83459 (Cherrybuck) rivets may be re-hit after installation, provided:

(1) The proper die and a bucking bar are used when re-hitting the rivet.

(2) The rivet manufactured head and rivet driven button dimensions remain within specification limits.

(3) The rivets do not crack.

(4) The rivet is not installed with automatic riveting machines, or hand-held electromagnetic riveters.

TABLE XI - BACR15FV AND NASM14218 RIVET REPLACEMENT

RIVET DIAMETER (INCH)	THINNEST ALLOWABLE SKIN IN WHICH RIVET MAY BE INSTALLED (INCH) FL 1
5/32	0.050
3/16	0.056
7/32	0.072
1/4	0.081
9/32	0.090

FL 1 In certain cased a hot-bonded doubler may be treated as part of the skin thickness for the purpose of determining coutersink depth for rivet rework. Consult Liaison Engineering for applicability.

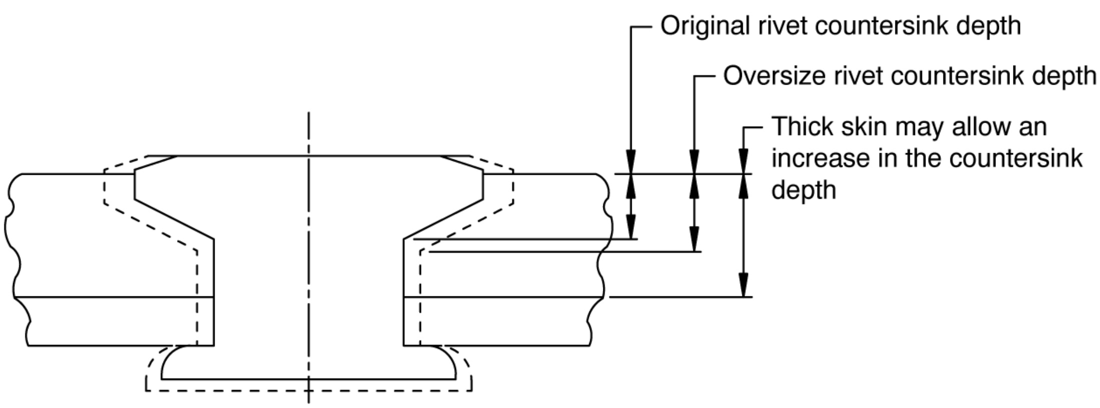


FIGURE 5 - NASM14218 AND BACR15FV RIVET IN THICK SKIN

8.1.9 REWORK (Continued)

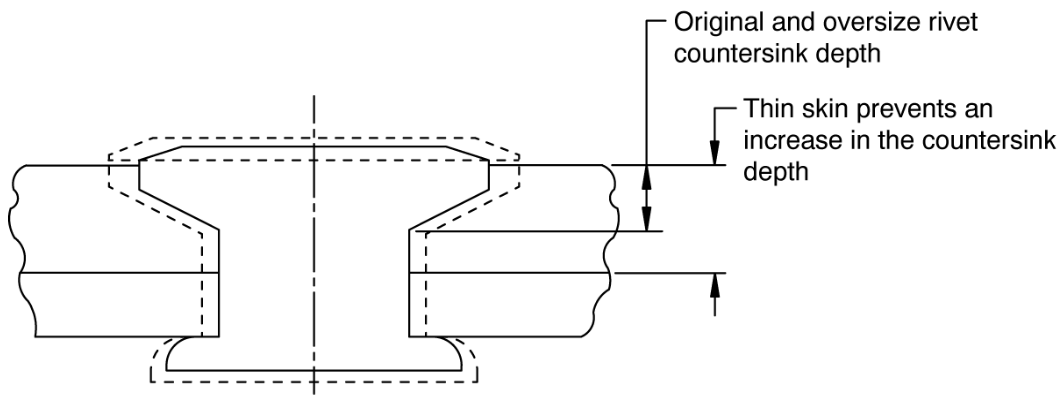


FIGURE 6 - NASM14218 AND BACR15FV RIVET IN THIN SKIN

8.2 IN-PROCESS VERIFICATION

a. Hydraulic-squeeze installation equipment used to install BACR15FV, NASM14218, NAS1097, BACR15CE and BACR15GF rivets in the exterior fuselage skin structure, to the requirements of this specification, shall produce test coupons in gages and alloys which are representative of the structure being riveted. Test coupons shall have dimensions in accordance with the developed and documented program installation plan, or 65-76487, as appropriate, shall contain a minimum of four open holes and five installed rivets, and shall be prepared and made available to Quality Assurance for buyoff at the following minimum intervals:

(1) Each time a new panel or assembly of panels is started.

(2) Each time a drill or countersink cutter is changed.

b. On machines with automatic tool changing devices, test coupons for all applicable diameters may be processed prior to panel loading.

c. Quality Assurance may require more or less frequent inspection of test coupons at the discretion of Quality Assurance Management. However, statistical data is required as a basis for more or less frequent inspection.

d. Test coupon requirements and inspection requirements shall be included in the Manufacturing and Operating plans.

e. Test coupons for BACR15GF, BACR15CE and NAS1097 rivets shall be inspected to the requirements of Table I and this specification.

f. Test coupons for BACR15GF rivets installed with a 100° CSK/CB cutter shall be inspected to the requirements of Table IV, Table VI, and this specification.

g. Test coupons for BACR15FV and NASM14218 rivets shall be inspected to the requirements of Table III, Table V, and this specification.

8.3 STATISTICAL PROCESS CONTROL (SPC) OF RIVET INSTALLATIONS

a. Manual installation and installation with hydraulic-squeeze equipment of countersunk rivets in exterior fuselage skin structure shall be accomplished through the use of a documented SPC or in-process verification plan approved by Manufacturing and Quality Assurance.

b. SPC or in-process verification requirements shall be included in a process control document approved by Manufacturing and Quality Assurance. See Section 8.1.1.

c. Rivets used on non-aerodynamic surfaces or for nutplate attachment are exempt from this specification requirement.

d. Subcontractors shall submit their SPC or in-process verification plan to Boeing Material & Process Technology for approval.

8.4 PERSONNEL REQUIREMENTS

Manufacturing and Inspection personnel shall meet the requirements of their locally owned Process Control Document and demonstrate their capability prior to install or inspect BACR15FV and NASM14218 rivets, and BACR15GF rivets prepared using 100° CSK/CB countersink cutters, to the requirements of this specification. Demonstration of capability during training shall include the ability to correctly install and inspect rivets in a test coupon for automated equipment, or a test panel for hand installation. Personnel Certification to this capability shall not be required.

9 MAINTENANCE CONTROL

a. When installing the BACR15FV rivet with hydraulic-squeeze equipment, the upper anvil contact surface shall not exceed the rivet head diameter. The tool drawing shall require that the rivet standard number BACR15FV and the rivet shank diameter number appear on the upper anvil and/or upper anvil assembly. This number may appear as a portion of a larger tool number.

b. All hydraulic-squeeze equipment used to install NAS1097, BACR15CE, BACR15GF, BACR15FV, and NASM14218 rivets in the exterior fuselage skin structure, to the requirements of this specification, shall be checked to the requirements of this specification and D6-56617 after rework or overhaul when such rework affects the riveting process or installed rivet characteristics.

c. Machine alignment of hydraulic-squeeze equipment shall be checked quarterly, or in accordance with the schedule established for equipment certified to the requirements of D6-56617, or on a schedule that is based on data from a statistical preventive maintenance plan developed for the machine. Subcontractors shall submit their machine maintenance plans to Material & Process Technology for approval.

10 QUALITY CONTROL

a. Assure that the requirements of this specification are met by monitoring the process and examining the end-items in accordance with established quality assurance provisions.

10 QUALITY CONTROL (Continued)

b. The inspection criteria given in Table XII shall be adhered to. Use a sampling plan in accordance with BSS7286 using the Inspection Reliability Requirement (IRR) values specified herein. Inspection lot definition shall be in accordance with the operating company's Quality Assurance Provisions.

c. Rework shall be documented as required by the operating company's is Quality Assurance Manual Provisions.

d. An equivalent Quality Assurance approved plan which provides assurance that the requirements of this process specification are being met may be used in place of sampling in accordance with BSS7286 for any individual IRR in Table XII.

e. Test coupons produced for automated installation shall be inspected to the requirements of this specification.

f. Quality Assurance shall verify that manual installation and installation by hydraulic-squeeze equipment of countersunk rivets in exterior fuselage skin structure is accomplished through the use of an SPC or in-process verification plan in accordance with Section 8.3.

g. Quality Assurance shall verify that facilities used to perform this process are certified to the applicable requirements of this specification. Verify that only personnel who have demonstrated their capability during training, in accordance with Section 8.4, perform this process.

TABLE XII - INSPECTION RELIABILITY REQUIREMENT (IRR)

CHARACTERISTIC	IRR (PERCENT) FL 1	LOT SIZE	SAMPLE SIZE
1. Holes a) Diameter b) Surface Defects c) Angularity d) Countersinking - Dimpling	90	Up to 200	6
		201 and Up	7
2. Shaving and Flushness 3. Driven Rivet Buttons a) Dimensions b) Cracks c) Deformed Buttons 4. Gaps, Shanking, Part Separation 5. Damage to Sheet and Rivets	90	Up to 200	6
		201 and Up	7
6. All Other Engineering Requirements	80	Up to 100	3
		101 and Up	4

FL 1 Acceptance requires zero defects in the sample.

11 REQUIREMENTS

11.1 FLUSHNESS

11.1.1 HEAD FLUSHNESS

a. Unless otherwise noted on the Engineering drawing, countersunk rivet heads on nonaerodynamic surfaces shall be flush within the following limits:

(1) BACR15GF, BACR15CE and NAS1097 rivets shall be flush within +0.005, -0.000 inch.

(2) BACR15FV and NASM14218 rivets shall be flush within +0.006, +0.001 inch.

(3) All other countersunk rivets shall be flush within +0.010, -0.000 inch.

b. Unless otherwise noted on the Engineering drawing, countersunk rivet heads on aerodynamic surfaces shall be flush within the following limits:

(1) BACR15FV and NASM14218 rivets shall be flush within +0.006, +0.001 inch.

(2) BACR15GF rivet nominal flushness shall be +0.003, with a lower limit of +0.001 and upper limit of +0.006. This is a key characteristic.

(3) All other countersunk rivets shall be flush within +0.002, -0.001 inch.

c. BACR15FV, BACR15GF, NAS1097, BACR15CE, and NASM14218 rivet heads shall not be shaved or modified to meet flushness requirements, except as allowed for rework of BACR15FV rivets in accordance with Section 8.1.9e..

Rivets other than BACR15FV, NAS1097, BACR15GF, BACR15CE and NASM14218 shall be installed above flush and may be shaved when necessary to meet flushness requirements.

d. Flushness of the BACR15CE and NAS1097 rivet is measured from the edge (flat area) of the rivet head to the adjacent skin.

e. Flushness of the BACR15FV and NASM14218 rivet is measured from the highest point on the rivet head to the adjacent skin, 0.040 inch maximum from the rivet head. Measure flushness with ST1416-5, ST1416-9, or equivalent gage.

f. When measured manually, the head flushness of BACR15GF rivets shall be measured with a flat disk stylus to the highest point of the installed rivet head. Measure flushness with the ST1416-9, WST132512, or equivalent gage.

11.1.2 SHAVING

Shaving of rivet heads and driven buttons shall be in accordance with the applicable portion of Section 8.1.6, Section 8.1.7 and Section 8.1.9 unless otherwise noted on the Engineering drawing. Unless specified by the Engineering drawing, shaved rivets do not require subsequent finish.

11.2 DRIVEN BUTTONS AND MANUFACTURED HEADS

11.2.1 STANDARD RIVET DRIVEN BUTTON DIMENSIONS

Driven rivet button dimensions shall be in accordance with applicable portions of Section 8.1.7 and as follows, unless otherwise noted on the Engineering Drawing.

The ST1396AA or equivalent gage is recommended for measurement of driven button dimensions.

a. When installing BACN10FM, BACN10LK, BACN10YF and nutplates of like design, the minimum driven button diameter shall be in accordance with Table XIII and the rivet driven button height shall be a minimum of 0.010 inch above the surface of the nutplate. Slight clinching of the rivet is acceptable provided the minimum rivet driven button dimensions are met (see Figure 7). Care should be taken to prevent excessive deformation to the flared holes and resulting restriction of the float.

b. When rivets are installed with a single impact tool (electromagnetic riveter) that has a centering/location die, driven buttons having raised centers will result. This shape (illustrated in Table XVIII) is acceptable provided the button raised center meets the requirements of Table XVIII. Single impact driving is prohibited for rivets with an undriven shear strength of 40 ksi or greater [i.e., 7050-T73, 2024-T4 (not "-W" condition), and 2017-T4 (since 2017 shear strength could exceed 40 ksi for a given lot)].

c. Driven rivet button dimensions for all solid CRES and nickel-copper alloy rivets (for example, NASM20613, NASM20615 and NASM20427) shall be in accordance with Table XIV.

d. Driven rivet button dimensions for NASM83459 titanium (Cherrybuck) rivets shall be in accordance with Table XV.

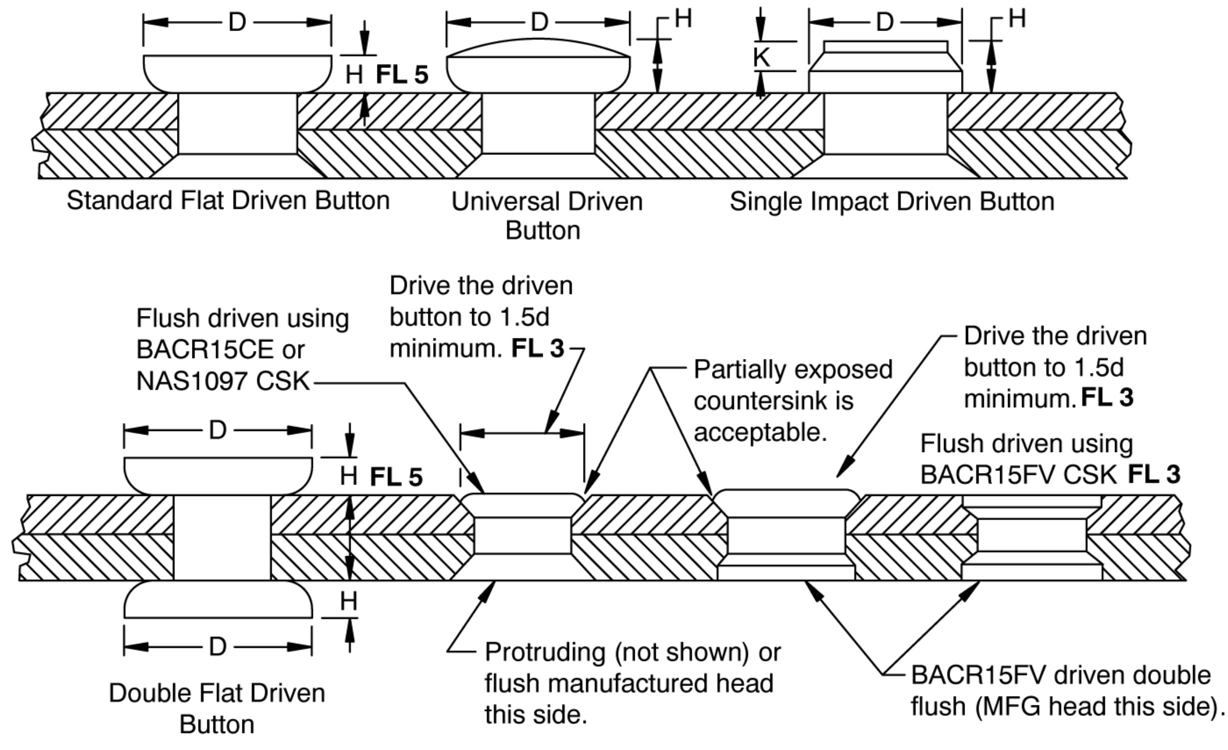
e. Required protrusion before driving tubular rivets shall be in accordance with Table XVI.

f. Installed tubular rivets shall have the bucked head turned back tightly against the bottom sheet. See Figure 8.

g. Radial cracks in the bucked head of an installed tubular rivet are acceptable. See Figure 9.

11.2.1 STANDARD RIVET DRIVEN BUTTON DIMENSIONS (Continued)

TABLE XIII - DRIVEN RIVET BUTTON DIMENSIONS (INCH) FOR ALUMINUM RIVETS



RIVET DIA. CODE	NOMINAL RIVET DIA.	D MINIMUM DRIVEN RIVET BUTTON DIAMETER			H DRIVEN RIVET BUTTON THICKNESS OR HEIGHT			
		ALL RIVETS EXCEPT AS NOTED	BACR15GF, AND HAND DRIVEN 7050 ALUMINUM ALLOY RIVETS	MACHINE DRIVEN AND SQUEEZED 7050 ALUMINUM ALLOY RIVETS	BACR15GF, BACR15CE, BACR15DS, BACR15FV NASM14218, NAS1097 MINIMUM	ALL OTHER RIVETS MIN.	FL 4 MAXIMUM RECOM-MENDED	K MAXIMUM
		1.3d	1.4d	1.5d				
2	1/16	0.081	0.088	0.094	0.025	0.025	0.040	- - -
3	3/32	0.122	0.131	0.141	0.038	0.038	0.060	- - -
4	1/8	0.165	0.175	0.188	0.050	0.050	0.080	0.030
5	5/32	0.203	0.219	0.234	0.050	0.062	0.100	0.037
6	3/16	0.245	0.264	0.282	0.060	0.075	0.120	0.045
61 FL 1 FL 2	13/64	- - -	0.284	0.305	0.065	- - -	0.130	0.049
7 FL 1	7/32	0.285	0.311	0.333	0.070	0.085	0.140	0.052
71 FL 1	15/64	- - -	0.332	0.351	0.075	- - -	0.150	0.056

11.2.1 STANDARD RIVET DRIVEN BUTTON DIMENSIONS (Continued)

TABLE XIII - DRIVEN RIVET BUTTON DIMENSIONS (INCH) FOR ALUMINUM RIVETS (Continued)

RIVET DIA. CODE	NOMINAL RIVET DIA.	D MINIMUM DRIVEN RIVET BUTTON DIAMETER			H DRIVEN RIVET BUTTON THICKNESS OR HEIGHT			
8	1/4	0.325	0.350	0.375	0.080	0.100	0.160	0.060
81 FL 1 FL 2	17/64	- - -	0.372	0.398	0.085	- - -	0.170	0.064
9 FL 1	9/32	0.365	0.397	0.425	0.090	0.110	0.180	0.067
91 FL 1	19/64	- - -	0.417	0.445	0.105	- - -	0.190	0.071
10	5/16	0.406	0.438	0.465	0.125	0.125	0.200	0.075
11 FL 1	11/32	0.450	0.481	0.515	0.135	0.135	0.210	0.082
12	3/8	0.488	0.525	0.562	0.150	0.150	0.210	0.090
13 FL 1	13/32	0.530	0.569	0.609	0.165	0.165	0.215	0.097

FL 1 See Section 8.1.9, Rework.

FL 2 The D minimum driven rivet button diameter for BACR15GF61 shall be 0.305 (1.5d). The D minimum driven rivet button diameter for BACR15GF81 shall be 0.398 (1.5d).

FL 3 The driven button of a BACR15FV rivet may be driven flush into an BACR15FV, NAS1097, NASM14218, BACR15GF or BACR15CE countersink. Use the driven rivet button requirements which are appropriate for the selected countersink.

FL 4 Exceeding button height adds unnecessary weight and should be avoided.

FL 5 Apply H dimensional requirements to the lowest point of driven button when rivet driven buttons are driven on a slope or curved surface.

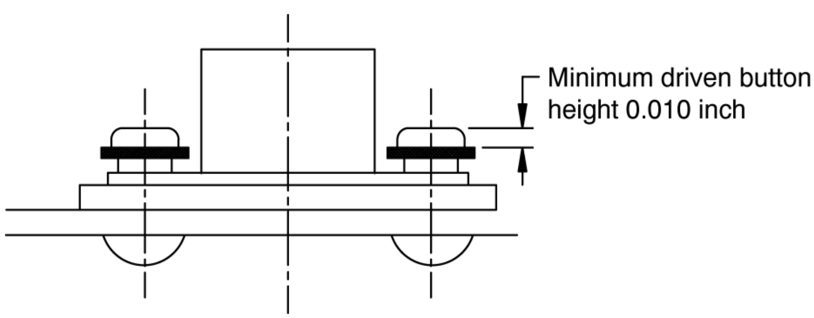
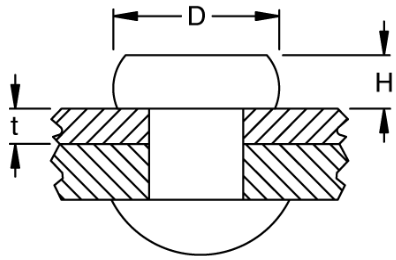


FIGURE 7 - INSTALLATION OF EYELET-TYPE NUTPLATES

11.2.1 STANDARD RIVET DRIVEN BUTTON DIMENSIONS (Continued)

TABLE XIV - DRIVEN RIVET BUTTON DIMENSIONS (INCH) FOR CRES AND NICKEL-COPPER ALLOY RIVETS (DOES NOT INCLUDE TUBULAR RIVETS)



NOMINAL RIVET DIAMETER	THICKNESS (t) OF MATERIAL ADJACENT TO RIVET BUTTON	DRIVEN RIVET BUTTON THICKNESS OR HEIGHT H MIN.	DRIVEN RIVET BUTTON DIAMETER D MIN.
3/32	0.016 to 0.050 0.050 and above	0.023 0.038	0.113 0.122
1/8	0.016 to 0.050 0.051 and above	0.030 0.050	0.150 0.163
5/32	0.016 to 0.050 0.051 and above	0.035 0.062	0.180 0.203
3/16	0.016 to 0.050 0.051 and above	0.040 0.075	0.222 0.244
1/4	0.016 to 0.050 0.051 and above	0.050 0.087	0.275 0.325

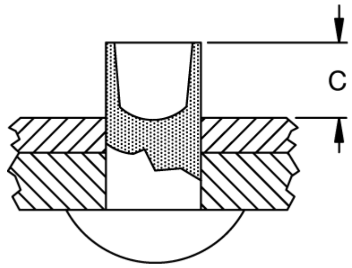
TABLE XV - DRIVEN RIVET BUTTON DIMENSIONS (INCH) FOR NASM83459 TITANIUM (CHERRYBUCK) RIVETS

NOMINAL RIVET DIAMETER	BUTTON THICKNESS OR HEIGHT FL 1	MINIMUM BUTTON DIAMETER
5/32 3/16	0.056 0.065	0.213 0.246
1/4 5/16	0.085 0.106	0.325 0.408
3/8	0.128	0.488

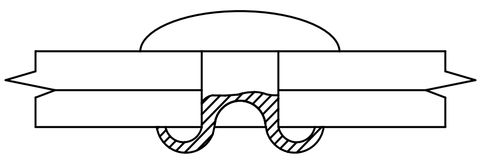
FL 1 Minimum upset height = 0.34d.

11.2.1 STANDARD RIVET DRIVEN BUTTON DIMENSIONS (Continued)

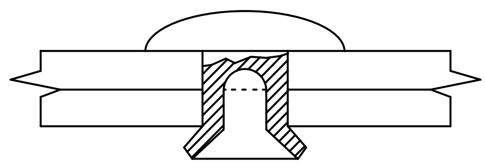
TABLE XVI - REQUIRED PROTRUSION FOR TUBULAR RIVETS



RIVET DIAMETER (INCH)	C (INCH)	
	MINIMUM	MAXIMUM
3/32	0.04	0.07
1/8	0.06	0.09
5/32	0.08	0.11
3/16	0.11	0.14

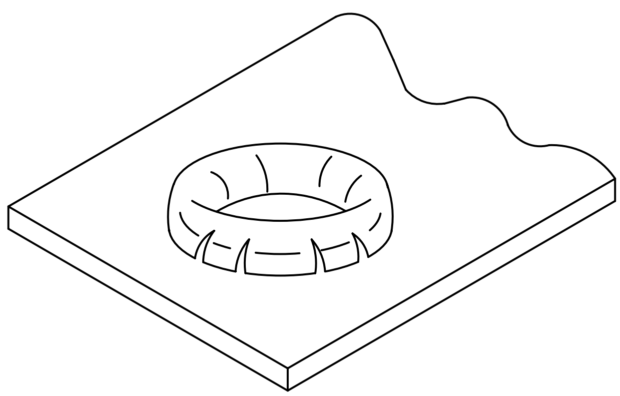


Acceptable



Not Acceptable
Rivet too Short

FIGURE 8 - TUBULAR RIVET INSTALLATIONS



Acceptable

FIGURE 9 - RADIAL CRACK ILLUSTRATION

11.2.2 MANUFACTURED HEADS

a. A deformation of the manufactured head caused by the mating cupped die is acceptable provided there are no sharp discontinuities in the deformed surface (see Figure 10 and Figure 11).

b. Damage to the structural surface, resulting from the use of a rivet die with too large a radius as shown in Figure 12 is a rejectable condition.

c. Cracks in manufactured heads after installation are not allowed.

d. Flattened head height of the manufactured protruding head (BACR15BB, NASM20470 or BACR15FT) shall not be less than those values of Table XVII.

e. The manufactured head of BACR15FV and NASM14218 rivets shall show evidence of having been driven. See Section 8.1.8 for tape use information.

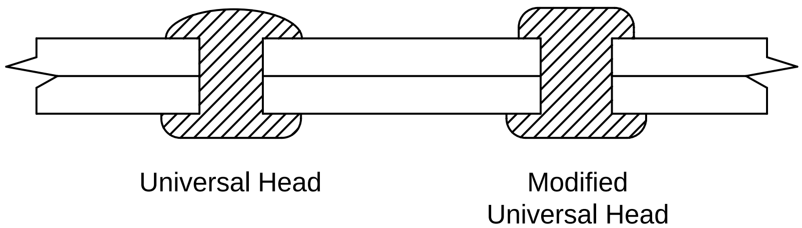


FIGURE 10 - ACCEPTABLE DEFORMATION OF THE MANUFACTURED HEAD

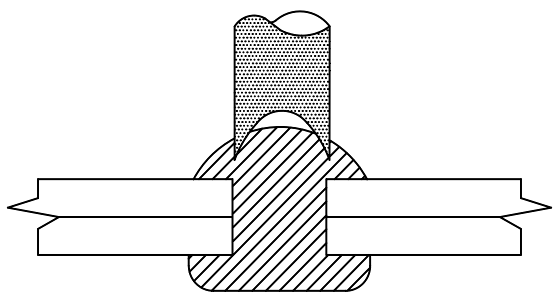


FIGURE 11 - DEFORMATION NOT ACCEPTABLE

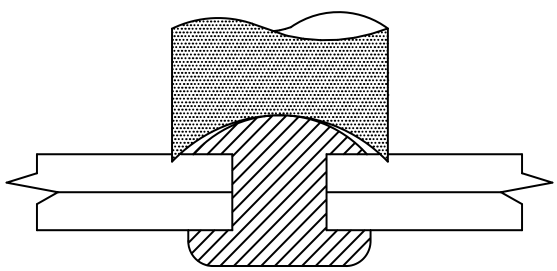
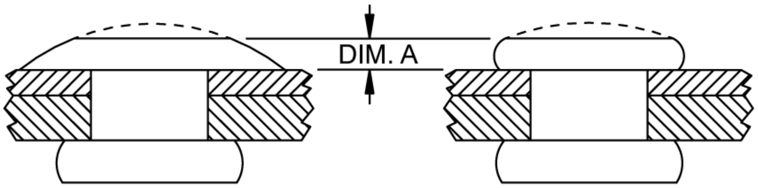


FIGURE 12 - OVERSIZED RIVET DIE

11.2.2 MANUFACTURED HEADS (Continued)

TABLE XVII - MINIMUM UNIVERSAL HEAD HEIGHT (INCH)



NOMINAL RIVET DIAMETER	MINIMUM UNIVERSAL HEAD HEIGHT, DIM. A
1/8	0.030
5/32	0.040
3/16	0.050
7/32	0.050
1/4	0.060
9/32	0.060
5/16	0.080
11/32	0.080
3/8	0.100
13/32	0.100
7/16	0.120

FL 1 Universal head.

FL 2 Modified universal head.

11.2.3 DIAGONAL CRACKS

Solid rivet driven button diagonal cracks are defined to be those cracks that run at an angle (other than 90 degrees) to the flat surface of the top of the button. When diagonal cracks occur, riveting operations shall be investigated for possible errors in rivet heat-treat condition or rivet lengths. Diagonal cracks are unacceptable. See Section 8.1.9 Rework.

11.2.4 VERTICAL CRACKS

Solid rivet driven button vertical cracks and axial discontinuities run perpendicular to the flat surface of the top of the button. Vertical cracks are the result of overheating during heat treatment. Axial discontinuities are from a pre-existing condition (laps, seams, inclusions, tool marks, etc.) in the rivet material. Vertical cracks are unacceptable. Axial discontinuities that exceed a depth of 0.004 inch are unacceptable. See Section 8.1.9 Rework.

11.2.5 DEFORMED DRIVEN BUTTON

a. Clinched or bent-over buttons are not acceptable if the hole or deburred area is visible. See Figure 13.

11.2.5 DEFORMED DRIVEN BUTTON (Continued)

b. Bell-shaped rivet buttons denote incomplete driving and are not acceptable. See Figure 14.

c. For rivets driven in automatic riveting machines, limits on the maximum allowable height of marking of driven heads, caused by wear of the locating pin, are shown in Table XVIII. This mark shall not extend below the top of the driven rivet button. See Section 11.2.6 for Driven Button Identification Marking requirements.

d. Out-of-round rivet driven buttons should be judged by the minimum diameter. Out-of-round and tipped rivet driven buttons are acceptable if the button dimensions meet those listed in Table XIII, Table XIV and Table XV. Cut heads are unacceptable.

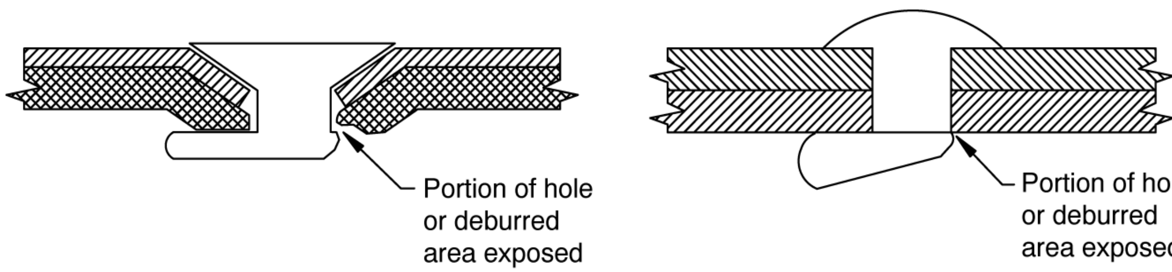


FIGURE 13 - CLINCHED RIVETS NOT ACCEPTABLE

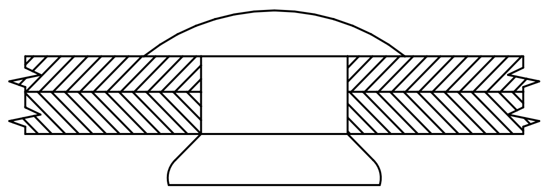
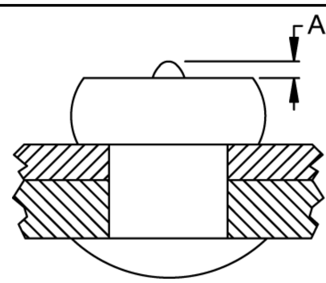


FIGURE 14 - BELL-SHAPED RIVETS NOT ACCEPTABLE

TABLE XVIII - MAXIMUM ALLOWABLE HEIGHT OF MARKS ON SINGLE IMPACT DRIVEN RIVET BUTTON



RIVET DIAMETER (INCH)	A (INCH)
3/32	0.005
1/8	0.010
5/32	0.015
3/16	0.020

11.2.6 DRIVEN BUTTON IDENTIFICATION MARKING

To assist inspectors to differentiate between hand driven and portable squeeze tool driven 7050 alloy rivets, the squeeze tool die may form an identifying mark, letter Z, on the driven rivet button. This mark shall conform to the height limitations of Table XVIII.

11.3 GAPPED AND SHANKED RIVETS

11.3.1 GAPPED RIVETS

a. There shall be no measurable gap between the driven button of an installed solid shank rivet and the countersink hole, when driven flush.

b. Gaps around BACR15FV or NASM14218 rivet heads are not acceptable. A BACR15FV or NASM14218 rivet head gap is defined as a space between the rivet head and the counterbore in the sheet. A gap does not include the shallow groove formed by the intersection of the rivet head periphery and the edge of the sheet counterbore or edge of slightly displaced skin material ("volcano"). See Figure 15.

c. The manufactured heads of all protruding head fasteners shall seat such that a 0.003-inch-thick shim as shown in Figure 16 does not enter between the structure and the manufactured head. Figure 17 illustrates the checking method.

d. The manufactured heads of all flush head fasteners except BACR15FV and NASM14218 rivets shall seat such that the radius tip of a 0.003-inch-thick shim as shown in Figure 16 cannot be inserted between the fastener head and the countersink when inspected as shown in Figure 20.

e. Gaps under driven buttons are acceptable provided a 0.003-inch shim as shown in Figure 16 cannot be inserted between the structure and the driven button, when inspected as shown in Figure 19. See also Section 11.2.5.

f. On tube assemblies, a gap under the manufactured head of all protruding fasteners or driven button (due to the curvature of the tube) is acceptable if the contact between the manufactured head and driven button, in a direction parallel to the tube axis, meets the protruding head requirements of Section 11.3.1c.. and Section 11.3.1e.. See Figure 18.

11.3.1 GAPPED RIVETS (Continued)

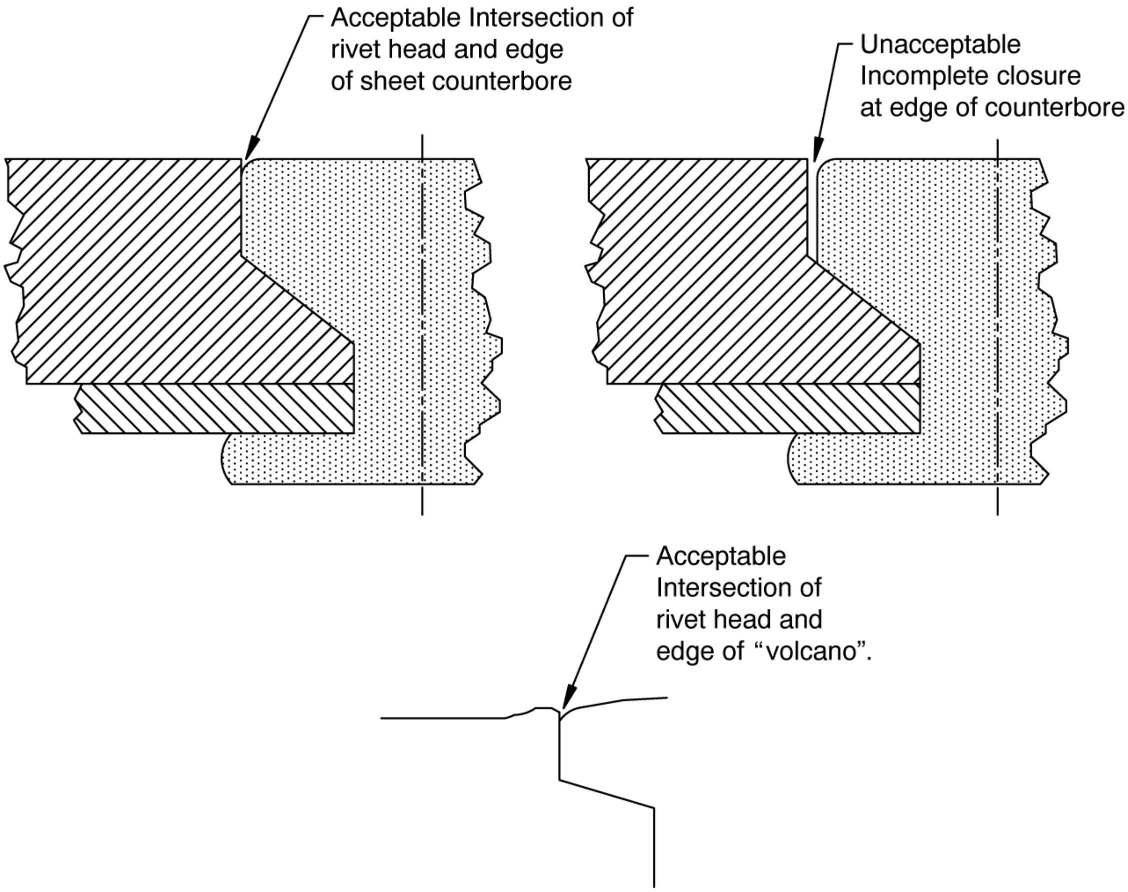


FIGURE 15 - BACR15FV OR NASM14218 HEAD GAPS

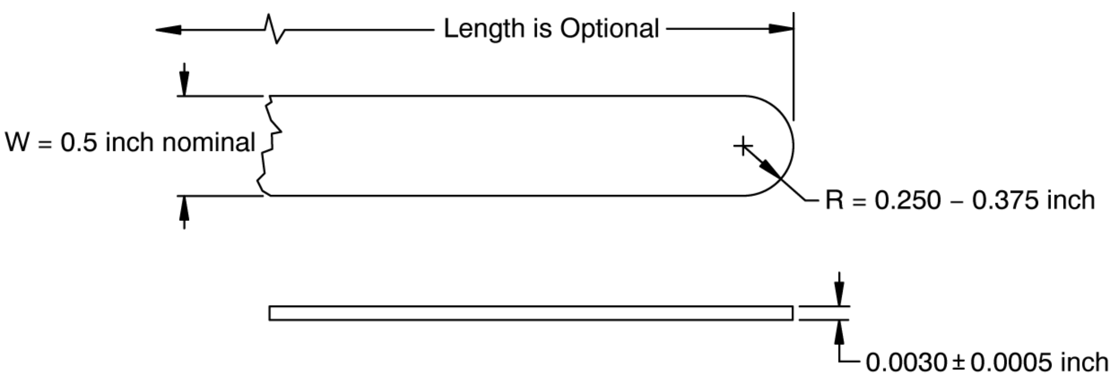


FIGURE 16 - GAP INSPECTION SHIM

11.3.1 GAPPED RIVETS (Continued)

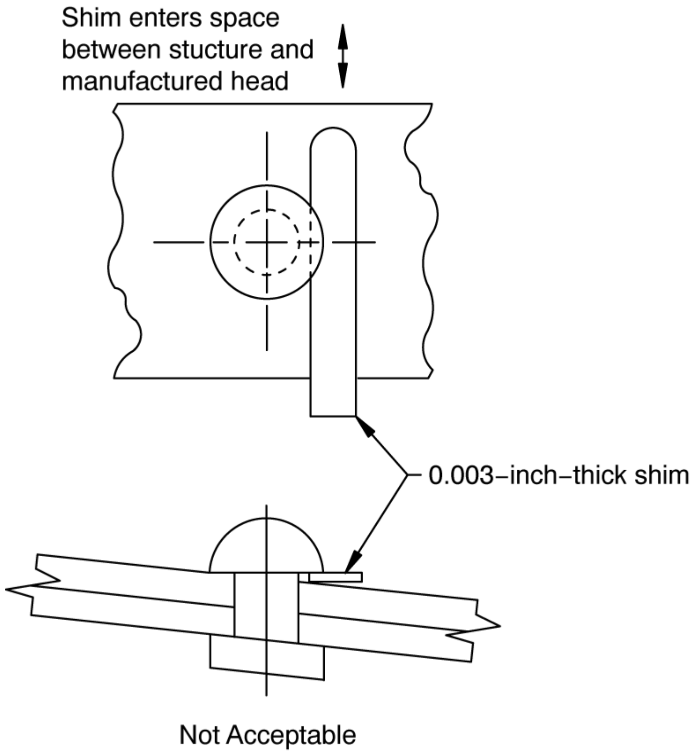


FIGURE 17 - PROTRUDING HEAD GAP CHECKING METHOD

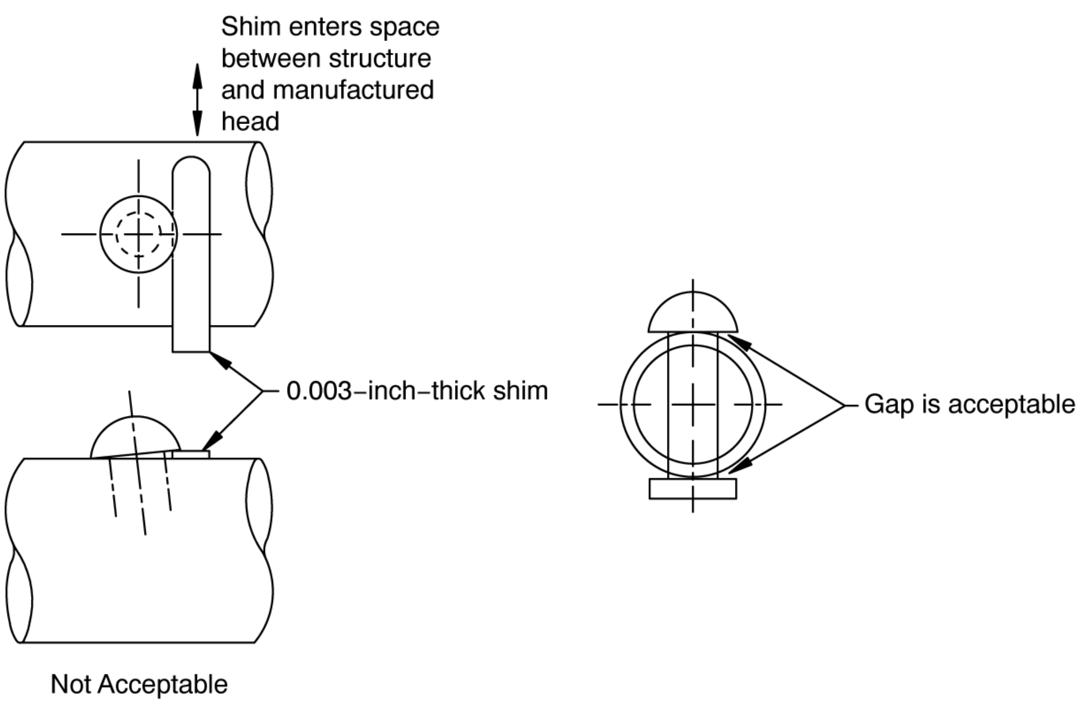


FIGURE 18 - PROTRUDING HEAD GAP CHECKING METHOD, TUBE ASSEMBLIES

11.3.1 GAPPED RIVETS (Continued)

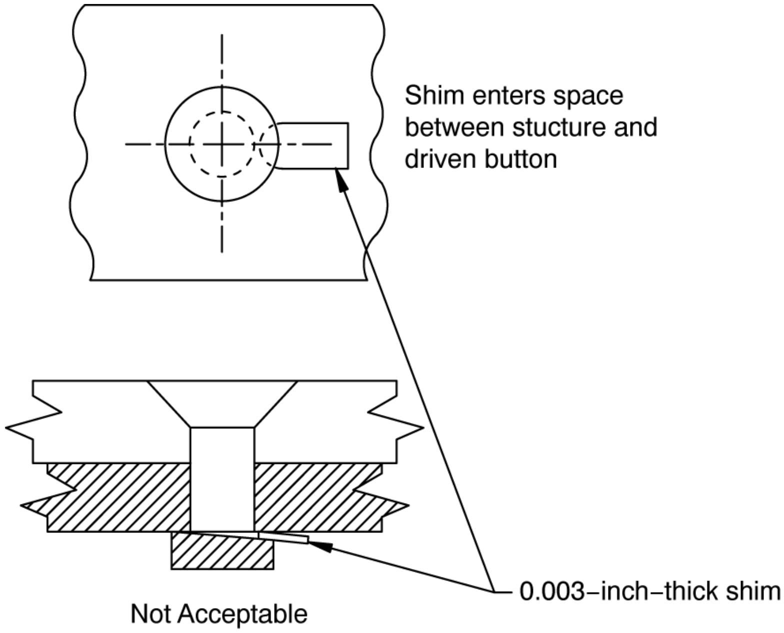


FIGURE 19 - DRIVEN BUTTON GAP CHECKING METHOD

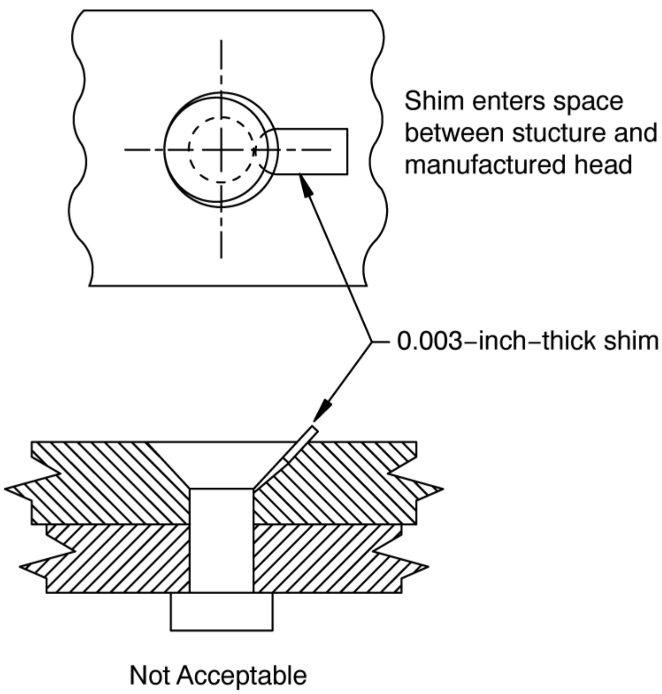


FIGURE 20 - FLUSH HEAD GAP CHECKING METHOD (EXCEPT BACR15FV AND NASM14218)

11.3.2 SHANKED RIVETS - PART SEPARATION AND SEALANT APPLICATION

a. Part separation at the rivet shank shall not exceed 0.015 inch for the rivet adjacent to intermediate material as illustrated in Figure 21.

b. Part separation at the rivet shank shall not exceed 0.015 inch for the rivet adjacent to intermediate material when tapered fillers are used (see Figure 22).

c. Part separation at the rivet shank shall not exceed 0.015 inch for a rivet in the joggle area when both parts of the structure are joggled (see Figure 23).

d. Part separation at the shank of rivets used only for the purpose of attaching nutplates and gang channel nuts shall not exceed 0.004 inch.

e. Part separation in fay seal areas between structure, fillers, shims, etc., may be 0.004 inch maximum, if the gap is completely filled with sealant in accordance with BAC5000 or Engineering drawing call out, and provided the gap does not extend to the rivet shank (See Section 11.3.2h. also).

f. Except in Section 11.3.2a.. through Section 11.3.2e.., part separation at the rivet shank shall not exceed 0.002 inch for stackups where the thinnest member is 0.050 inch or less and 0.003 inch where the thinnest member is greater than 0.050 inch.

g. Where no sealant is used, there shall be no measurable gap at the fastener shank at interfaces between structure, shims, fillers, et cetera. If a 0.003-inch-thick shim, as shown in Figure 16, can be slid between the parts and contact the rivet shank, the rivet shall be rejected for shanking except as specifically permitted in Section 11.3.2a.. through Section 11.3.2f..

h. Where sealant is used, all gaps shall be completely filled with sealant.

NOTE: Use caution in checking for gaps in sealant areas. If sealant is removed from the gap areas, the voids shall be refilled with fresh sealant or the fastener shall be replaced and the new fastener shall be installed with fresh sealant.

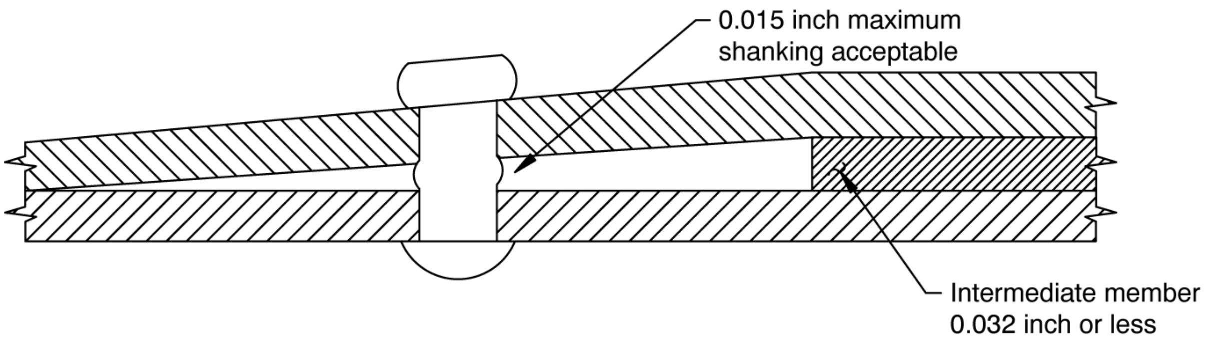


FIGURE 21 - PART SEPARATION WITHOUT FILLER

11.3.2 SHANKED RIVETS - PART SEPARATION AND SEALANT APPLICATION (Continued)

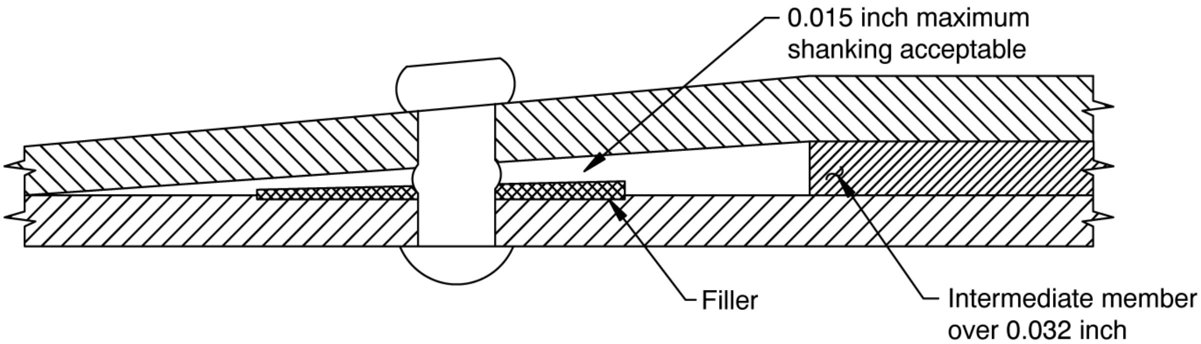


FIGURE 22 - PART SEPARATION WITH FILLER

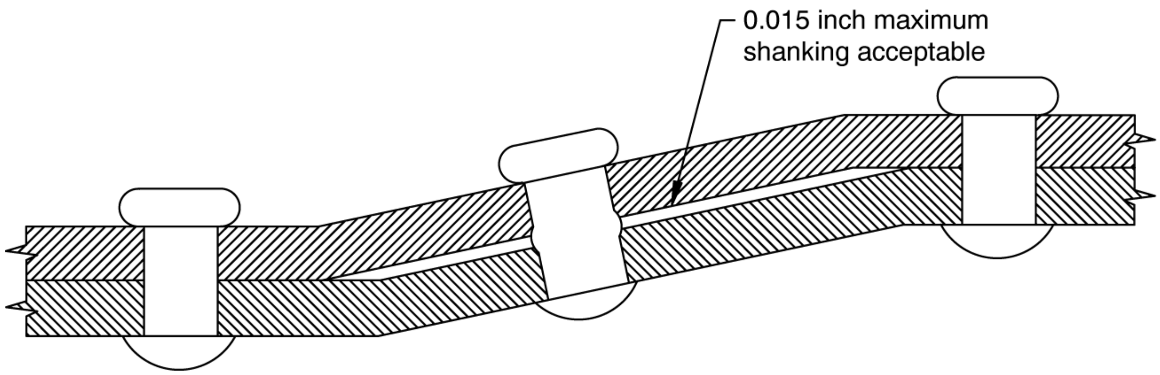


FIGURE 23 - PART SEPARATION IN JOGGLE

12 TEST METHODS

Not applicable to this specification.

13 QUALIFICATION

Not applicable to this specification.